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電子物理學系
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National Yang Ming Chiao Tung University
Master Thesis

二維材料於純量與向量渦旋光激發下之激子躍遷理論研究
Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

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二維材料於純量與向量渦旋光激發下之激子躍遷理論研究

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摘　要

光的自旋角動量於 1936 年首次於 Beth 的實驗中被量測，而結構光所攜帶的軌道角動量則於 1992 年由 Allen 等人透過拉蓋爾−高斯 (Laguerre−Gaussian) 光束於實驗中驗證。近年研究指出，在非近軸條件下，光場中可能出現自旋與軌道角動量之間的耦合。然而，僅在近軸近似下，光束才能同時具備明確的自旋與軌道角動量。最近的研究已指出，

~~自旋−軌道交互作用出現於近軸近似下蓋爾−高斯光束中微弱但非零的縱向分量之相位結構中。~~
~~vortex beams)，並證明其對應的自旋−軌道交互作用項可延續並保留於縱向場分量的振幅之中。~~
增加 SOC 重要性

~~為了提供觀測此自旋−軌道交互作用特徵的可行途徑，二維過渡金屬二硫族化物 (transition metal dichalcogenides) 成為研究光與物質交互作用的理想平台之一，這主要歸因於其在可見光至近~~
~~excitons) 因可透過光場的縱向分量與光耦合，特別適合用於探測此向量渦旋光中自旋−軌道交互作~~
~~bright excitons）與灰激子（gray excitons）皆可透過不同機制，利用向量渦旋光的相對相位調控，實~~

最後，本研究延伸先前對谷混合激子波包的探討，分析激子內部波函數中的相位結構，使得任意相位無法被允許提出並影響波包形貌。

~~雖然對於谷激子波包，相位結構在一定近似下仍可分離，但對於谷混合激子而言，問題更為複~~
激子相位對拓樸的重要性

關鍵字：純量與向量渦漩光、二維過度金屬二流族化合物、光的自旋-軌道交互作用、谷混合亮激子、灰激子、躍遷速率、波包

Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

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Abstract

The spin angular momentum (SAM) of light was established in 1936, while the orbital angular momentum (OAM) carried by structured light was later demonstrated in 1992 using Laguerre–Gaussian (LG) beams. Recent studies have shown that coupling between SAM and OAM can occur in non-paraxial (strongly focused) light. However, only within the paraxial regime can a light beam carry simultaneously well-defined SAM and OAM. Previous studies have identified SAM and OAM interaction (SOI) in the phase of the small but finite longitudinal component of paraxial beams. Here, we extend these studies by superposing two LG beams to form vector vortex beams (VVBs), and show that the corresponding SOI term is preserved in the amplitude of the longitudinal field.

To provide a practical route for observing this SOI feature in VVBs, two-dimensional transition metal dichalcogenides (TMDs) have emerged as promising platforms for studying light–matter interactions, owing to their finite direct bandgap in the visible to near–infrared spectral range and their strong excitonic properties. In particular, gray excitons (GXs), which couple to light through the out–of–plane (longitudinal) field component, are well suited for this purpose. Moreover, the transition dipole moment of GXs is approximately isotropic in momentum space, allowing the resulting light–matter interaction to directly reflect the SOI of VVBs. This study finds that both valley-mixed bright excitons (BXs) and gray excitons can be selectively excited via different mechanisms by exploiting the relative phase control of VVBs.

Furthermore, we extend previous studies on the realization of valley-mixed exciton wave packets by explicitly extracting the phase structure embedded in the internal exciton wave function, such that common factors cannot be trivially removed as a global phase. While the phase term can be approximately separated for valley exciton wave packets, the situation is more involved for valley-mixed excitons. In this case, not only phase extraction is required, but also a more careful construction of the exciton wave function is necessary to avoid arbitrary phase factors that can significantly alter the resulting wave-packet profile.

Keywords: Scalar and vector vortex beams, two dimensional transition metal dichalcogenides, spin-orbit interaction of light, valley-mixed bright excitons, gray exciton, transition rate, wave packet

Contents

Figure Captions

Chapter 1 Introduction

1.1 Exciton in Transition Metal Dichalcogenide (TMD) Monolayer

Conventional semiconductors such as diamond, silicon [1], and aluminum arsenide are not suitable for the selective initialization, manipulation, and readout of valley states. This is because they possess multiple inequivalent valleys in their band structures that do not couple distinctly to external optical fields. The situation changed with the discovery of graphene in 2004 [2], which triggered rapid progress in research on two-dimensional (2D) materials. Graphene has a hexagonal honeycomb lattice that places two inequivalent valleys at the K and K' points of the hexagonal Brillouin zone. Nevertheless, the absence of a bandgap in graphene allows electrons to be thermally excited between valleys, making valley polarization unstable at room temperature.

In contrast, transition-metal dichalcogenide (TMD) monolayers share a similar honeycomb lattice but possess a finite direct bandgap (see Figure 1.1) on the order of $\sim$ 1–2 eV, which lies in the visible to near-infrared spectral range. A TMD monolayer consists of two group-VIA chalcogen atoms, such as sulfur (S) or selenium (Se), with a group-VIB transition-metal atom, such as molybdenum (Mo) or tungsten (W), located between them.

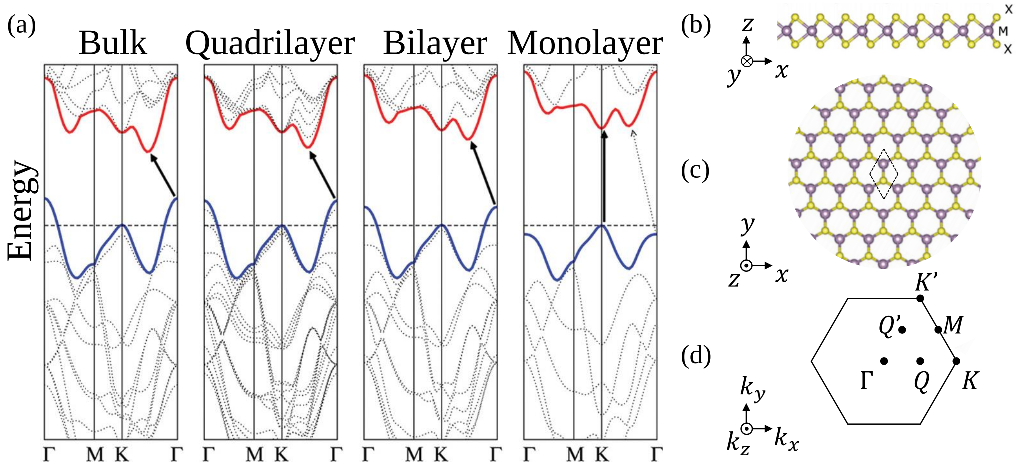

Figure 1.1 : (a) Calculated band structures of MoS_2 with different layers, reproduced from Ref. [3]. (b) Side view and (c) top view of transition-metal dichalcogenide (TMD) monolayer with the unit cell framed by the dashed line, reproduced from Ref. [4]. (d) The first Brillouin zone of TMD monolayer in the recipical space and the selected high symetry points

Under optical excitation, electrons in a TMD monolayer can be promoted from the valence band to the conduction band near the K and K' valleys, leaving holes in the valence band. Due to the strong Coulomb interaction in atomically thin materials with weak dielectric screening, the electron–hole pair can bind together to form a quasiparticle known as an exciton. Figure 1.2 shows that center-of-mass (COM) momentum of exciton is related to momentum of light $\boldsymbol{Q}_c$ with incident angle θ through $\boldsymbol{Q} = \boldsymbol{Q}_c \sin\theta$, which gives the idea of the light cone.

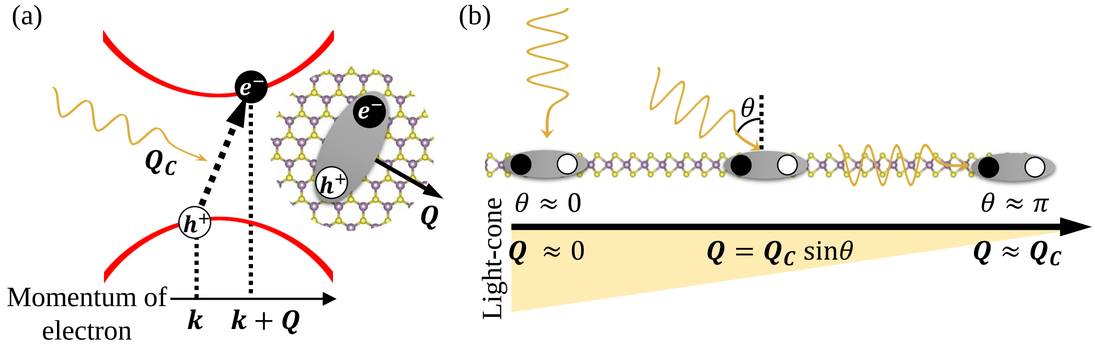

Figure 1.2：(a) Illustration of a photon carrying momentum $\boldsymbol{Q}_c$ exciting (leaving) an electron (hole) with crystal momentum $\boldsymbol{k} + \boldsymbol{Q}$ $(-\boldsymbol{k})$, forming an exciton with center-of-mass (COM) momentum $\boldsymbol{Q}$. (b) Schematic of ideal of light cone and relation between COM momentum $\boldsymbol{Q}$ and $\boldsymbol{Q}_c$

Moreover, the lack of inversion symmetry in the lattice, together with strong spin–orbit coupling from the heavy transition-metal atoms, gives rise to pronounced spin splitting of the electronic bands. At the K/K' valleys, this splitting is typically on the order of a few meV in the conduction band and several hundreds of meV in the valence band, as illustrated in Figure 1.3(a).

The same symmetry considerations also lead to valley-dependent optical selection rules. As shown in Figure 1.3(b), excitons can be classified according to both momentum and spin selection rules. Only excitons with center-of-mass (COM) momentum $\boldsymbol{Q}$ within the light cone,

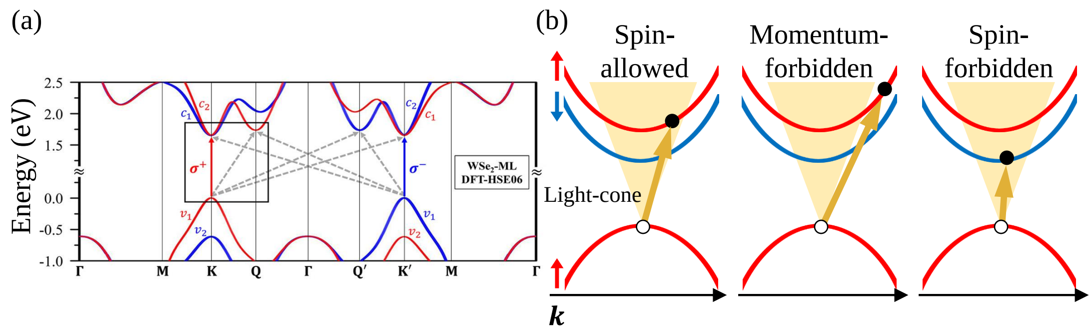

Figure 1.3：(a) Calculated band structure of WSe_2, showing spin-up (red) and spin-down (blue) states, together with a schematic illustration of valley-selective optical transitions at the K and K' valleys, which couple to right- and left-circularly polarized light, respectively, reproduced from Ref. [5]. (b) Schematic configurations of excitonic states, including spin-allowed bright excitons (BXs), momentum-forbidden dark excitons (DXs), and spin-forbidden DXs.

i.e., $|\boldsymbol{Q}| \lesssim Q_c$, and with antiparallel spin configurations can recombine radiatively and are classified as bright excitons (BXs). In contrast, excitons that lie outside the light cone or violate the spin selection rules remain optically inactive and are therefore classified as dark excitons (DXs).

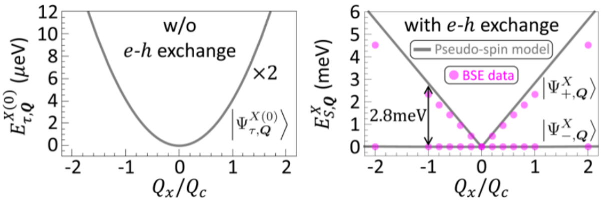

Figure 1.4 : The lowest valley-degenerate (left) and valley-split (right) exciton bands in MoS_2 along the Q_x direction, reproduced from Ref. [6].

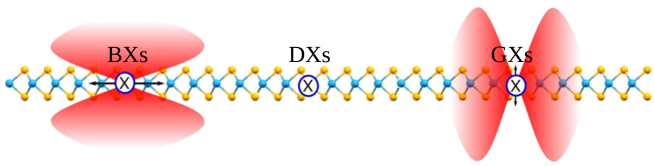

Figure 1.5 : Illustration of the radiation patterns of BXs, DXs, and GXs in neutral WSe_2, along with the corresponding optical excitation indicated by arrows (reproduced from Ref. [7]).

The finer band structure further enriches the classification of quasiparticles. In particular, the lowest bright exciton band, which is initially twofold degenerate, splits into two linearly dispersing branches when electron–hole exchange interactions are included within a pseudospin model. These branches are commonly referred to as valley-mixed excitons [6].

In W-based TMDs, an additional excitonic state known as the gray exciton (GX) can also emerge. As shown in Figure 1.5, GXs emit light polarized perpendicular to the two-dimensional plane (out-of-plane direction). Compared with BXs and DXs, GXs exhibit relatively long lifetimes—typically an order of magnitude longer than BXs (BX $\sim$ 10 ps) [7]—while retaining weak but finite optical activity. These properties make GXs particularly useful for studying exciton dynamics and light–matter interactions, and highlight the rich excitonic physics in two-dimensional TMDs.

1.2 Spin and Orbital Angular Momentum (SAM and OAM) of Laguerre–Gaussian (LG) beam

Following the theoretical work of Poynting in 1909 [8] and the experimental demonstration by Beth in 1936 [9], it has been established that each photon in circularly polarized light with finite dimensions [10–12] carries a spin angular momentum (SAM) of $S = \sigma\hbar = \pm\hbar$ along the propagation direction, where σ denotes the photon helicity. In Beth's experiment, the torque exerted on a suspended half-wave plate by circularly polarized light was measured, demonstrating that spin angular momentum can be transferred to a birefringent medium during propagation (see Figure 1.6(a)).

In 1992, Allen *et al.* showed that a laser beam in a Laguerre–Gaussian (LG) mode possesses a well-defined orbital angular momentum (OAM), also directed along the propagation axis, with $L = \ell\hbar$ per photon, where $\ell \in \mathbb{Z}$ is the azimuthal mode index, commonly referred to as the topological charge [10]. In their experiment, an input Hermite–Gaussian mode, which carries zero OAM, was converted into an LG mode with nonzero OAM. This mode conversion subsequently induced a measurable torque on cylindrical optical elements (see Figure 1.6(b, c)), establishing that light can also carry OAM associated with the spatial structure of its wavefront. The helical phase structure associated with a quantized amount of $\ell\hbar$ of OAM is illustrated in Figure 1.7, while intensified charge-coupled device (ICCD) images show the characteristic doughnut-shaped intensity distribution.

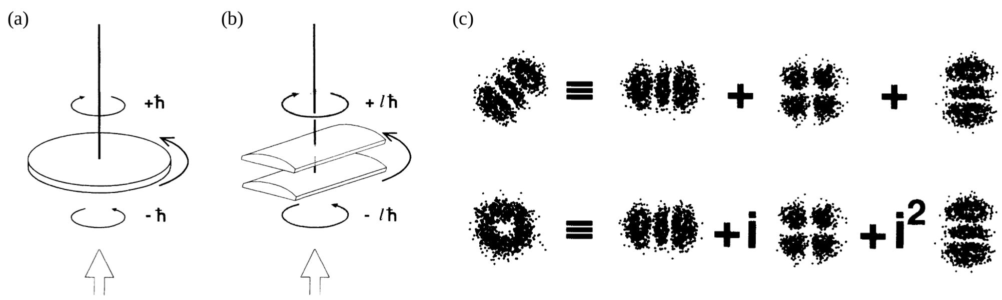

Figure 1.6：(a) Illustration of the conversion between right-handed and left-handed circularly polarized light generating a torque on a suspended birefringent half-wave plate. (b) Illustration of transforming a Laguerre–Gaussian (LG) mode carrying OAM $\ell\hbar$ per photon into a mode with $-\ell\hbar$ per photon, which produces a mechanical torque on suspended cylindrical lenses. (c) Demonstration of generating an LG beam with nonzero ℓ (left side of bottom row) by introducing a phase shift when converting Hermite–Gaussian modes with zero ℓ (left side of top row). All reproduced from Ref. [10].

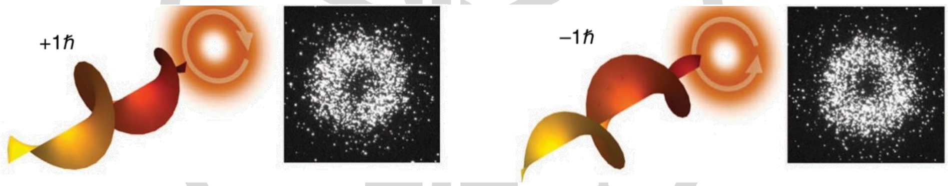

Figure 1.7：Illustration of twisted light with a helical wavefront, where each photon carries $\ell\hbar$ of orbital angular momentum. The intensified charge-coupled device camera images show the characteristic doughnut-shaped intensity profile. Reproduced from Ref. [13].

Since then, a wide range of applications of light carrying OAM has been developed. Early studies focused on angular momentum transfer from light to matter, where objects experience mechanical torque due to OAM transfer, resulting in rotational motion [14–16]. Later developments have extended to optical communications [17–19], quantum key distribution [20, 21], astronomy [22], data storage [23, 24], enhanced imaging sensitivity [25, 26], and optical tweezers [14, 27]. These advances motivate investigation of how structured light carrying orbital angular momentum interacts with electronic excitations in low-dimensional materials.

1.3 SAM and OAM coupling in LG beam

In free space, coupling between SAM and OAM has been shown to occur in non-paraxial (tightly focused) light. Along the propagation direction, SAM can be partially converted into

OAM as the focusing angle increases [28]. This effect can be understood by considering the reduction of the spin component along the beam axis induced by a focusing lens (see Figure 1.8).

The initial spin s_0 is aligned with the propagation direction. After focusing, the spin is no longer parallel to the beam axis; instead, the final spin s_1 is tilted, so only its longitudinal component s_z contributes to the beam' s spin angular momentum. Since the maximum spin angular momentum is $s\hbar$ per photon, the total spin is reduced. However, total angular momentum is conserved during focusing. Consequently, this reduction in SAM is compensated by orbital angular momentum generation, leading to a transfer from SAM to OAM [29].

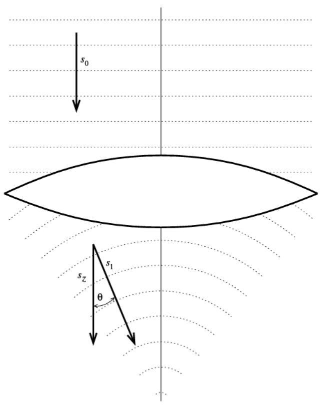

Figure 1.8：Illustration of a beam focused by a lens, showing the variation of SAM before and after the lens along the propagation direction. Reproduced from Ref. [29].

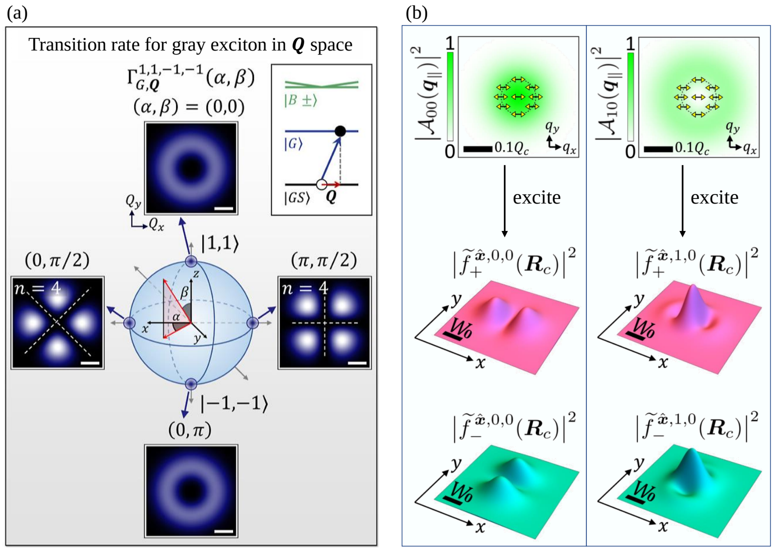

Figure 1.9：(a) Transition-rate distribution for gray excitons in momentum space, reproduced from Ref. [30]. (b) Distribution of the incident light and the excited BX wave-packet states, reproduced from Ref. [6].

However, it should be emphasized that the simple proportionality between spin and orbital angular momentum with the optical helicity σ and topological charge ℓ holds strictly only within the paraxial approximation [31, 32], which often neglects the longitudinal component. However, it is precisely this longitudinal component that can couple to GXs. More importantly, this neglected component already contains two distinct types of spin−orbit interaction (SOI).

In 2024, Oscar Javier Gomez Sanchez *et al.* demonstrated that a type of SOI originating from the phase of the longitudinal component induces momentum-dependent transition rates of GXs through the superposition of two twisted-light beams, forming the vector vortex beams (VVBs) (see Figure 1.9(a)) [30]. However, the combined contribution of SAM and OAM was not fully clarified in their formulation. Meanwhile, in 2022, Guan-Hao Peng *et al.* investigated the photoexcitation of valley-mixed BXs by LG beams within the pseudospin model, suggesting a connection to excitonic topology [6] (see Figure 1.9(b)). Nevertheless, the global phase of the valley-mixed excitons may still influence the excited wave-packet profile.

In this work, we extend the studies of Oscar Javier Gomez Sanchez *et al.* and Guan-Hao Peng *et al.* to investigate the interplay between spin and orbital angular momentum of light in determining transition rates and wave-packet distributions of excitonic states within SI units.

Chapter 2 SAM and OAM interaction (SOI) of LG Beam

2.1 Gauge freedom

Although electromagnetic fields are the directly observable physical quantities, structured light beams such as Laguerre–Gaussian (LG) beams are often formulated in terms of the scalar and vector potentials. However, the potentials possess gauge freedom, leading to different forms of the wave equations under different gauge conditions. Importantly, the derivation of LG beams relies on the paraxial approximation, whose implementation is not completely independent of the chosen gauge.

In this section, we examine how gauge choice affects the formulation of wave equations. We begin by introducing the wave equations under different gauge conditions, and then analyze the compatibility of the paraxial approximation. This discussion clarifies why certain gauges are more suitable than others for describing paraxial optical fields.

2.1.1 Gauge Freedom in Wave Equations

Maxwell's equations describe how the **electric field** $\boldsymbol{E}(\boldsymbol{r}, t)$ and the **magnetic field** $\boldsymbol{B}(\boldsymbol{r}, t)$ evolve in space and time. These equations form the fundamental basis for describing electromagnetic waves, including Laguerre–Gaussian beams. In **free space**, the charge density ρ and current density $\mathbf{J}$ vanish. In this case, Maxwell's equations reduce to four homogeneous differential equations involving the vacuum permittivity ϵ_0, the vacuum permeability μ_0, and the speed of light in vacuum $c = 1/\sqrt{\mu_0 \epsilon_0}$.

Within this electrodynamics framework, Maxwell's equations can be rewritten as two coupled wave equations for the **scalar and vector potentials**, $\phi(\boldsymbol{r}, t)$ and $\boldsymbol{A}(\boldsymbol{r}, t)$, derived in appendix A.1 as

$$\nabla^2 \phi(\boldsymbol{r}, t) + \frac{\partial}{\partial t} (\boldsymbol{\nabla} \cdot \boldsymbol{A}(\boldsymbol{r}, t)) = 0, \tag{2.1}$$

$$\nabla^2 \boldsymbol{A}(\boldsymbol{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \boldsymbol{A}(\boldsymbol{r}, t) - \boldsymbol{\nabla} \left(\boldsymbol{\nabla} \cdot \boldsymbol{A}(\boldsymbol{r}, t) + \frac{1}{c^2} \frac{\partial}{\partial t} \phi(\boldsymbol{r}, t) \right) = 0. \tag{2.2}$$

These equations contain the same physical information as the original Maxwell equations, but are expressed in terms of the scalar and vector potentials.

Further simplification of Eqs. (2.1) and (2.2) can be achieved by exploiting the gauge freedom of the potentials. Below, we introduce two commonly used gauges: the **Coulomb gauge** and the **Lorenz gauge**. Superscripts C and L are used to distinguish quantities defined in the Coulomb and Lorenz gauges, respectively. The gauge conditions are given by

$$\text{Coulomb gauge condition}: \quad \boldsymbol{\nabla} \cdot \boldsymbol{A}^C(\boldsymbol{r}, t) = 0, \tag{2.3}$$

$$\text{Lorenz gauge condition}: \quad \boldsymbol{\nabla} \cdot \boldsymbol{A}^L(\boldsymbol{r}, t) = -\frac{1}{c^2}\frac{\partial}{\partial t}\phi^L(\boldsymbol{r}, t). \tag{2.4}$$

Both gauges simplify the wave equations. In the Coulomb gauge, they reduce to

$$\nabla^2 \phi^C(\boldsymbol{r}, t) = 0, \text{ and} \tag{2.5}$$

$$\nabla^2 \boldsymbol{A}^C(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\boldsymbol{A}^C(\boldsymbol{r}, t) = \frac{1}{c^2}\frac{\partial}{\partial t}\boldsymbol{\nabla}\phi^C(\boldsymbol{r}, t) = 0, \tag{2.6}$$

where the last equality follows from Eq. (2.5). In the Lorenz gauge, Eqs. (2.1) and (2.2) become

$$\nabla^2 \phi^L(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi^L(\boldsymbol{r}, t) = 0, \text{ and} \tag{2.7}$$

$$\nabla^2 \boldsymbol{A}^L(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\boldsymbol{A}^L(\boldsymbol{r}, t) = 0. \tag{2.8}$$

Here, we introduce the **monochromatic wave** ansatz, in which the electromagnetic fields and potentials are expressed as $\boldsymbol{E}(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{E}(\boldsymbol{r})e^{-i\omega t}\right]$, $\boldsymbol{A}^C(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{A}^C(\boldsymbol{r})e^{-i\omega t}\right]$, $\boldsymbol{A}^L(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{A}^L(\boldsymbol{r})e^{-i\omega t}\right]$, and $\phi^L(\boldsymbol{r}, t) = \text{Re}\left[\phi^L(\boldsymbol{r})e^{-i\omega t}\right]$, where ω is the angular frequency of light. Substituting these expressions into Eq. (A.1.6) from Maxwell's equations and applying the corresponding gauge conditions leads to

$$\boldsymbol{E}(\boldsymbol{r}) = i\omega \boldsymbol{A}^C(\boldsymbol{r}) = i\omega\left(\boldsymbol{A}^L(\boldsymbol{r}) + \frac{\boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A}^L(\boldsymbol{r})\right)}{q_0^2}\right). \tag{2.9}$$

Here $q_0 = \omega/c$ denotes the wavenumber of light in free space. Under the monochromatic-wave ansatz, electric field $\boldsymbol{E}(\boldsymbol{r}, t)$ and magnetic field $\boldsymbol{B}(\boldsymbol{r}, t)$ are related to potentials through Eq. (2.9) and Eq. (A.1.5)

2.1.2 Incompatibility of Gauge Freedom with the Paraxial Approximation

In principle, the vector potential formulation of LG beams can be derived starting from either Eq. (2.6) or Eq. (2.8). However, within Coulomb gauge, the directional constraint impose on the transversely polarized wave is ultimately incompatible with the standard paraxial Helmholtz formulation. For clarity, this incompatibility will be discussed in detail after deriving the paraxial Helmholtz equation within the Lorenz gauge framework [33].

The derivation starts from the scalar Helmholtz equation, which is obtained from Eq. (2.8) by imposing both the monochromatic ($\boldsymbol{A}^L(\boldsymbol{r}, t) = \text{Re}\left[\boldsymbol{A}^L(\boldsymbol{r})e^{-i\omega t}\right]$) and transverse wave ansätze ($\boldsymbol{A}^L(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}A^L(\boldsymbol{r})$, where we condsider polarization unit vector $\hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{\varepsilon}}_\sigma = \hat{\boldsymbol{x}} + i\sigma\hat{\boldsymbol{y}}$, $\sigma = \pm 1$ as right/left circularly polarized)(see Sec. A.2):

$$\nabla^2 A^L(\boldsymbol{r}) + q_0^2 A^L(\boldsymbol{r}) = 0. \tag{2.10}$$

The **paraxial approximation** [33–35] assumes that the rays have only small inclinations with respect to the optical axis in geometric optics. In wave optics, this corresponds to an angular spectrum dominated by plane-wave components whose wavevectors are close to the propagation direction. Here, we choose the propagation direction along the $\boldsymbol{z}$ axis, such that the dominant component of the wavevector $\boldsymbol{q}_0$ lies along $\hat{\boldsymbol{z}}$.

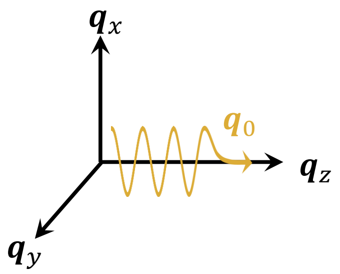

Figure 2.1 : Illustration of a dominant component of wavevector $\boldsymbol{q}_0$ lies along $\hat{\boldsymbol{z}}$

Under this approximation, we introduce the **paraxial wave** ansatz, $A^L(\boldsymbol{r}) = U(\boldsymbol{r})e^{iq_0 z}$, and substitute it into Eq. (2.10), which yields

$$\nabla_\parallel^2 U(\boldsymbol{r}) + \frac{\partial^2}{\partial z^2} U(\boldsymbol{r}) + 2iq_0 \frac{\partial}{\partial z} U(\boldsymbol{r}) = 0. \tag{2.11}$$

Here, $\nabla_\parallel^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ denotes the transverse Laplacian. The paraxial wave ansatz assumes that the **envelope function** $U(\boldsymbol{r})$ **varies slowly along the $\boldsymbol{z}$ direction** compared to the rapid phase variation of $e^{iq_0 z}$. Consequently, the paraxial approximation implies the following relations:

$$\left| \frac{\partial^2}{\partial z^2} U(\boldsymbol{r}) \right| \ll \left| \nabla_\parallel^2 U(\boldsymbol{r}) \right|, \tag{2.12}$$

$$\left| \frac{\partial^2}{\partial z^2} U(\boldsymbol{r}) \right| \ll q_0 \left| \frac{\partial}{\partial z} U(\boldsymbol{r}) \right|. \tag{2.13}$$

Neglecting the second-order derivative with respect to z then leads to the paraxial Helmholtz

equation

$$\nabla_\parallel^2 U(\boldsymbol{r}) + 2iq_0 \frac{\partial}{\partial z} U(\boldsymbol{r}) \approx 0. \tag{2.14}$$

We now clarify why a transversely polarized light ($\hat{\varepsilon}$ confined to the x–y plane) is not suitable in this derivation under the Coulomb gauge. For a transversely polarized wave, the Coulomb gauge condition gives

$$\begin{aligned} \boldsymbol{\nabla} \cdot \boldsymbol{A}^C(\boldsymbol{r}) &= \hat{\boldsymbol{\varepsilon}}_\sigma \cdot \boldsymbol{\nabla}_\parallel A^C(\boldsymbol{r}) && (2.15) \\ &= \left(\frac{\partial}{\partial x}\hat{\boldsymbol{x}} + i\sigma \frac{\partial}{\partial y}\hat{\boldsymbol{y}} \right) A^C(\boldsymbol{r}) = 0 && (2.16) \end{aligned}$$

where $\boldsymbol{\nabla}_\parallel = \frac{\partial}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial}{\partial y}\hat{\boldsymbol{y}}$. This condition imposes a directional constraint in the transverse plane on the vector potential and results in the transverse Laplaian of the potential become zero.

$$\nabla_\parallel^2 A^C(\boldsymbol{r}) = \left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y} \right) \left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y} \right) A^C(\boldsymbol{r}) = 0, \tag{2.17}$$

which in turn is incompatible with the approximation in Eq. (2.12) [33]. In contrast, a transversely polarized light do not have such constraint within Lorenz gauge. For Coulomb gauge, a nonzero longitudinal (z) component of the vector potential should be introduced, going beyond the simple transverse polarization. Therefore, the Lorenz gauge provides a more natural and consistent framework for deriving the paraxial Helmholtz equation and Laguerre–Gaussian beam solutions.

2.2 Manifestation of SOI in LG Beam

Within the paraxial approximation, the optical helicity σ and the topological charge ℓ are directly proportional to the spin and orbital angular momentum of light, respectively [31, 32]. This correspondence provides a convenient framework for analyzing the angular momentum properties of LG beams.

Moreover, when the vector potential formulation of LG beams are expressed in the Coulomb gauge, an SOI can emerge through the additional longitudinal component of the field, giving rise to terms proportional to $\sigma\ell$ and $\sigma + \ell$. Adopting the Coulomb gauge is further justified by the theory of light−matter interaction, where the vector potential is conventionally formulated within both the Coulomb gauge and the angular spectrum representation. Where in angular spectrum representation can we relate the in-plane wavevector to the vector in reciprocal space, which connect to theory of matter.

In the following section, we first introduce the LG beams solutions in both real-space and angular spectrum representations. We then perform a gauge transformation to reveal the SOI.

2.2.1 Solutions of LG Beams

Under the **long Rayleigh-range** approximation [36][1], the LG beam solutions of the paraxial Helmholtz equation, Eq. (2.14), can be expressed in cylindrical coordinates as

$$\boldsymbol{A}^{L}_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma A^{L}_{\ell,p}(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}_\sigma\, u_{\ell,p}(\boldsymbol{r}_\parallel)\, e^{iq_0 z}, \tag{2.18}$$

where the position vector is written as $\boldsymbol{r} = (\rho, \phi, z) = (\boldsymbol{r}_\parallel, z)$. Here $\boldsymbol{r}_\parallel$ denotes the in-plane position vector consisting of the radial coordinate ρ and the azimuthal angle ϕ. The integer $\ell \in \mathbb{Z}$ is the **topological charge**, and the integer $p \geq 0$ is the radial mode index. The vector $\boldsymbol{q}_0 = q_0\hat{\boldsymbol{z}}$ denotes the wavevector along the propagation direction. The envelope function is expressed as

$$u_{\ell,p}(\boldsymbol{r}_\parallel) = f_{\ell,p}(\rho)\, e^{i\ell\phi}, \tag{2.19}$$

where the radial function is $f_{\ell,p}(\rho) = A_0^{\mathrm{LG}}\sqrt{\frac{2p!}{\pi(|\ell|+p)!}}\, L_p^{|\ell|}\left(\frac{2\rho^2}{W_0^2}\right)\left(\frac{\sqrt{2}\rho}{W_0}\right)^{|\ell|}\exp\left(-\frac{\rho^2}{W_0^2}\right)$, which is given by [6, 30, 36] and A_0^{LG} is a constant amplitude. Here the generalized Laguerre polyno-

[1] For a light beam, the Rayleigh range is the propagation distance from the beam waist to the point where the beam cross-sectional area doubles (see Figure 2.2). It is given by $z_R = \frac{1}{2}q_0 W_0^2$, where W_0 is the beam waist. When the Rayleigh range is much larger than the characteristic propagation distance of interest, diffraction-induced beam spreading becomes negligible. In this limit, the beam waist can be treated as approximately constant, and the envelope function may be regarded as independent of z. This simplification is referred to as the long Rayleigh-range approximation.

mial is defined as $L_p^{|\ell|}(x) = \sum_{m=0}^{p}(-1)^m \frac{(|\ell|+p)!}{(p-m)!(|\ell|+m)!m!}x^m$, with $x = \frac{2\rho^2}{W_0^2}$. The parameter W_0 denotes the beam waist of the LG beam, while the cylindrical coordinates are related to the Cartesian coordinates through $\rho = \sqrt{x^2+y^2}$ and $\phi = \tan^{-1}(y/x)$. The geometric meaning of the beam waist and the Rayleigh range is illustrated in Figure 2.2.

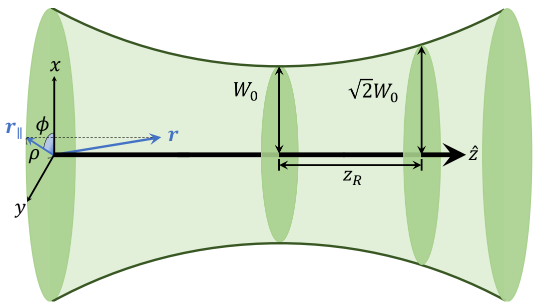

Figure 2.2：Illustration of the evolution of the beam radius $W(z)$ during propagation. The minimum beam radius W_0 occurs at the beam waist, and the Rayleigh range z_R is defined as the distance from the waist at which $W(z_R) = \sqrt{2}W_0$.

Focusing on the minimum cross-section area of the beam ($z = 0$), Figure 2.3 shows that as ℓ and/or p increase, $u_{\ell,p}(\boldsymbol{r}_\parallel)$ develops a broader doughnut-shaped intensity profile, accompanied by an increasing number of radial nodes along the radial direction. The phase varies continuously with the azimuthal angle ϕ, reflecting the helical wavefront associated with orbital angular momentum. In particular, the phase winds by $2\pi\ell$ around the beam center. The beam waist is chosen as $W_0 = 1.5\ \mu$m, which sets the spatial scale of the beam and determines the radial extent of the intensity distribution.

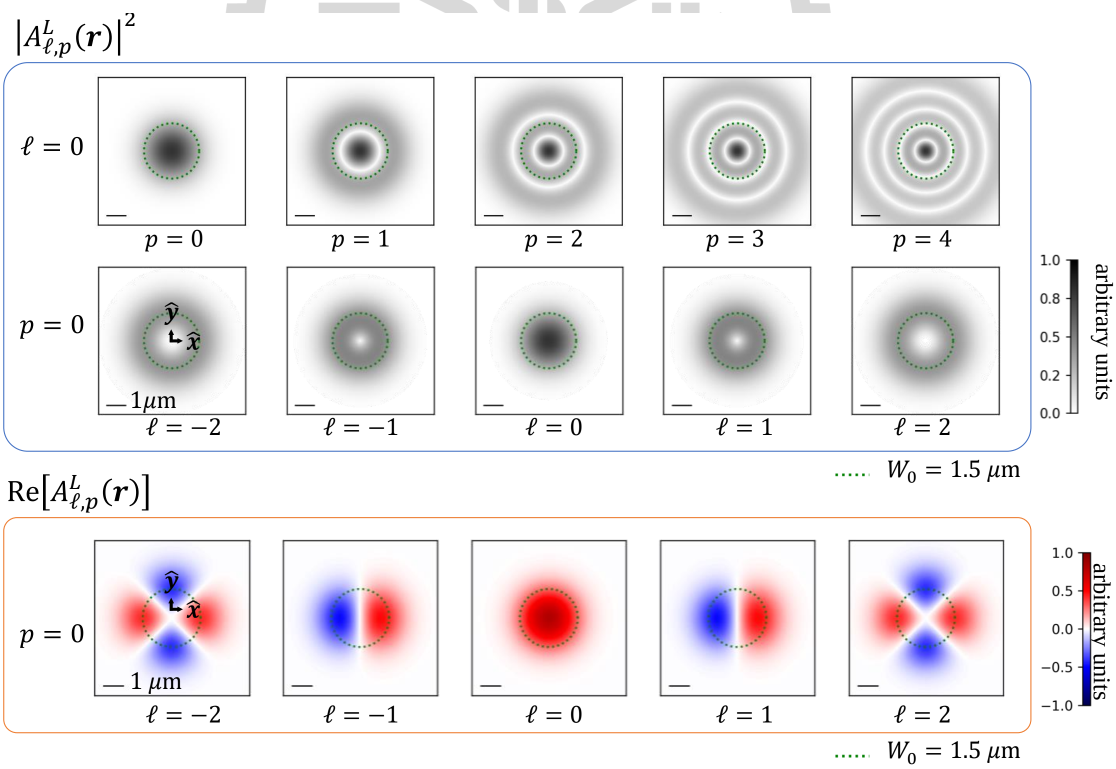

Figure 2.3：Squared magnitude (intensity) and real part distribution of $A^L_{\ell,p}(\boldsymbol{r})$ at $z = 0$. The green dotted line denotes the beam waist W_0. The scale bar indicates the length of 1 μm.

To derive the vector potential of LG beams in the angular spectrum representation, the potential is expanded **in a plane-wave basis** as

$$\boldsymbol{A}(\boldsymbol{r}) = \sum_{\boldsymbol{q}} \boldsymbol{\mathcal{A}}(\boldsymbol{q}) e^{i\boldsymbol{q}\cdot\boldsymbol{r}}, \tag{2.20}$$

where $\boldsymbol{\mathcal{A}}(\boldsymbol{q})$ is the complex amplitude associated with each plane-wave component $e^{i\boldsymbol{q}\cdot\boldsymbol{r}}$. The LG beams in Eq. (2.18) can be written in this form through a defined **two-dimensional Fourier transformation** and its corresponding inverse transformation,

$$u_{\ell,p}(\boldsymbol{r}_\parallel) = \frac{1}{(2\pi)^2} \int \mathrm{d}^2\boldsymbol{q}_\parallel \, F^{\mathrm{LG}}_{\ell,p}(\boldsymbol{q}_\parallel) \, e^{i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel}, \quad \longleftrightarrow \quad F^{\mathrm{LG}}_{\ell,p}(\boldsymbol{q}_\parallel) = \int \mathrm{d}^2\boldsymbol{r}_\parallel \, u_{\ell,p}(\boldsymbol{r}_\parallel) \, e^{-i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel}. \tag{2.21}$$

For applications to two-dimensional materials, the in-plane wavevector $\boldsymbol{q}_\parallel$ can be approximately identified with a reciprocal-space vector. Under this approximation, Eq. (2.21) can be discretized, and Eq. (2.20) becomes

$$\boldsymbol{\mathcal{A}}^L_{\sigma,\ell,p}(\boldsymbol{r}) = \sum_{\boldsymbol{q}_\parallel} \boldsymbol{\mathcal{A}}^L_{\sigma,\ell,p}(\boldsymbol{q}_\parallel) e^{i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel} e^{iq_0 z}. \tag{2.22}$$

Here the in-plane wavevector is written as $\boldsymbol{q}_\parallel = (q_\parallel, \phi_{\boldsymbol{q}_\parallel})$. The corresponding complex amplitude can be obtained analytically, as summarized in Table 2.1.

Table 2.1：Real space and angular spectrum representations of LG beams

Real space	**Angular spectrum**
Vector potential	
$\boldsymbol{A}^L_{\sigma,\ell,p}(\boldsymbol{r}) = \hat{\varepsilon}_\sigma A^L_{\ell,p}(\boldsymbol{r}) = \hat{\varepsilon}_\sigma \, u_{\ell,p}(\boldsymbol{r}_\parallel) \, e^{iq_0 z}$	$\boldsymbol{\mathcal{A}}^L_{\sigma,\ell,p}(\boldsymbol{q}_\parallel) = \hat{\varepsilon}_\sigma \mathcal{A}^L_{\ell,p}(\boldsymbol{q}_\parallel) = \hat{\varepsilon}_\sigma \frac{1}{\Omega} F^{\mathrm{LG}}_{\lvert\ell\rvert,p}(\boldsymbol{q}_\parallel)$
Envelope function	
$u_{\ell,p}(\boldsymbol{r}_\parallel) = f_{\ell,p}(\rho) e^{i\ell\phi}$	$\frac{1}{\Omega} F^{\mathrm{LG}}_{\ell,p}(\boldsymbol{q}_\parallel) = \tilde{F}_{\lvert\ell\rvert,p}(q_\parallel) e^{i\ell\phi_{\boldsymbol{q}_\parallel}}$
Solutions at $p=0$	
$f_{\ell,p=0}(\rho) = \frac{A_0^{\mathrm{LG}}}{\sqrt{(\lvert\ell\rvert)!\pi/2}} \left(\frac{\sqrt{2}\rho}{W_0}\right)^{\lvert\ell\rvert} \exp\left(-\frac{\rho^2}{W_0^2}\right)$	$\tilde{F}_{\lvert\ell\rvert,p=0}(q_\parallel) = \frac{A_0^{\mathrm{LG}} W_0^2 \, (-i)^{\lvert\ell\rvert} \, \pi}{\Omega\sqrt{(\lvert\ell\rvert)!\pi/2}} \left(\frac{q_\parallel W_0}{\sqrt{2}}\right)^{\lvert\ell\rvert} \exp\left(-\frac{q_\parallel^2 W_0^2}{4}\right)$

where $\tilde{F}_{\lvert\ell\rvert,p}(q_\parallel) = \frac{(-i)^{\lvert\ell\rvert}}{\Omega} 2\pi \, \mathcal{H}_{\lvert\ell\rvert}[f_{\ell,p}(\rho)]$, and $\mathcal{H}_{\lvert\ell\rvert}[f_{\ell,p}(\rho)] = \int_0^\infty \mathrm{d}\rho \;\; \rho \, f_{\ell,p}(\rho) \, J_{\lvert\ell\rvert}(q_\parallel \rho)$ is the **Hankel transform** of order $\lvert\ell\rvert$, where $J_{\lvert\ell\rvert}$ is the **Bessel function of the first kind**. Here Ω denotes the area of the two-dimensional system

Figure 2.4 shows that $\mathcal{A}^L_{\ell,p}(\boldsymbol{q}_\parallel)$ exhibits an intensity profile analogous to that of $A^L_{\ell,p}(\boldsymbol{r})$, reflecting the characteristic doughnut-shaped structure in angular spectrum. This correspondence follows directly from the Fourier transform relation between two representations of the LG beam. The doughnut-shaped distribution in angular spectrum indicates that the dominant in-plane wavevectors lie within a finite range determined by the beam waist W_0.

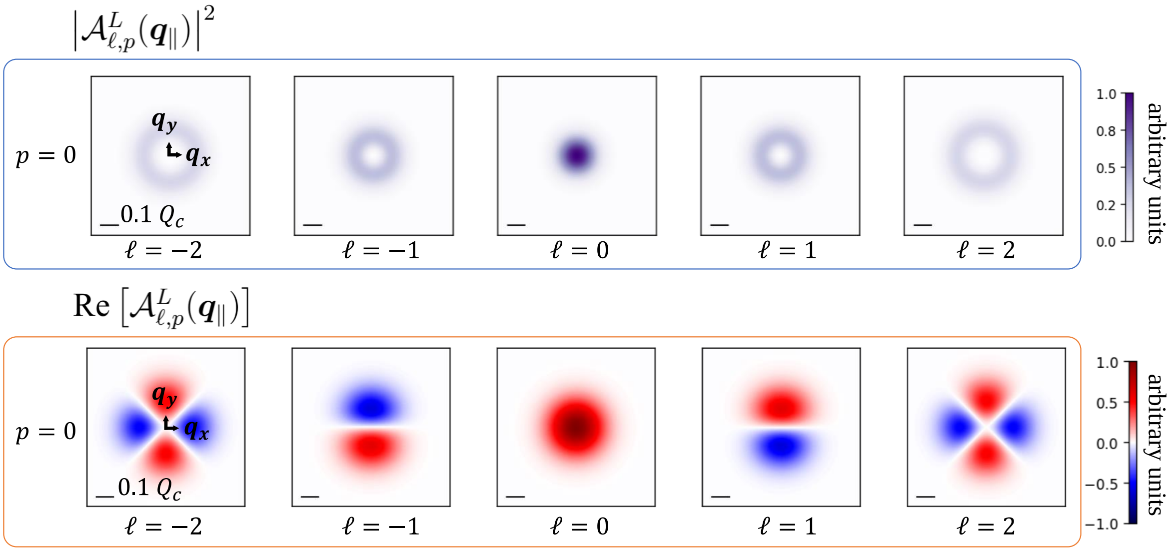

Figure 2.4：Squared magnitude (intensity) and real part distribution of $\mathcal{A}^L_{\ell,p=0}(\boldsymbol{q}_\parallel)$. The scale bar indicates the considered 0.1 times light cone range for WSe_2-monolayers, $Q_c = 8.6\ \mu\mathrm{m}^{-1}$.

2.2.2 Longitudinal Component of LG Beam：SOI effect

In the previous section, Eq. (2.9) was introduced to relate the vector potentials defined in the two gauges to the physical fields. Using this relation, we can perform gauge transformation by rewritten Eq. (2.9) as

$$\boldsymbol{A}^C_{\sigma,\ell,p}(\boldsymbol{r}) = \boldsymbol{A}^L_{\sigma,\ell,p}(\boldsymbol{r}) + \frac{\boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A}^L_{\sigma,\ell,p}(\boldsymbol{r})\right)}{q_0^2}. \tag{2.23}$$

To explicitly incorporate the SAM of light, the polarization unit vector is defined as $\hat{\boldsymbol{\varepsilon}} \equiv \hat{\boldsymbol{\varepsilon}}_\sigma = \frac{1}{\sqrt{2}}(\hat{\boldsymbol{x}}+i\sigma\hat{\boldsymbol{y}})$, where $\sigma = \pm 1$ denotes the spin of the photon. From now on we omit the superscript C to simplify the notations and the vector potential in real space under the Coulomb gauge is obtained as

$$\boldsymbol{A}_{\sigma,\ell,p}(\boldsymbol{r}) \equiv \boldsymbol{A}^C_{\sigma,\ell,p}(\boldsymbol{r}) = \tilde{A}^\parallel_{\ell,p}(\boldsymbol{r})\hat{\boldsymbol{\varepsilon}} + \tilde{A}^z_{\sigma,\ell,p}(\boldsymbol{r})\hat{\boldsymbol{z}}. \tag{2.24}$$

For LG beams in the fundamental radial mode $p = 0$

$$\begin{aligned}\tilde{A}^\parallel_{\ell,p}(\boldsymbol{r}) &\approx f_{\ell,p=0}(\rho)e^{i\ell\phi}e^{iq_0z}\\ \tilde{A}^z_{\sigma,\ell,p}(\boldsymbol{r}) &\approx \frac{i}{\sqrt{2}q_0\rho}\left((|\ell|+\sigma\ell) - \frac{2\rho^2}{W_0^2}\right)f_{\ell,p=0}(\rho)e^{ij\phi}e^{iq_0z}\end{aligned} \tag{2.25}$$

where

$$J \equiv \sigma + \ell$$

is the total angular momentum (TAM). In general, the second term in Eq. (2.23) introduces corrections to all spatial components of the vector potential. However, under the paraxial approximation, these corrections are suppressed. Therefore, we retain only the longitudinal (z) component as the leading correction.

The longitudinal component in Eq. (2.25) reveals two distinct types of SOI: a $\sigma\ell$ dependence in the amplitude and a J dependence in the phase. Figure 2.5 shows that the magnitude (absolute value) at $z = 0$ of this longitudinal component is significantly weaker than that of the transverse components. The $\sigma\ell$ dependence manifests as an asymmetry between cases with the same ℓ but opposite spin σ.

It is worth noting that the longitudinal term is weighted by a factor proportional to $(q_0 W_0^2)^{-1}$, which is inversely related to the Rayleigh range $(\frac{1}{2} q_0 W_0^2)$ [34] . This indicates that a smaller beam waist W_0 leads to a stronger longitudinal component, corresponding to a shorter Rayleigh range and tighter focusing, as shown in the bottom row of Figure 2.5. The beam waist W_0 is chosen to be 1.5 μm and 0.5 μm, representing a typical paraxial beam and a case at the boundary of the paraxial regime, respectively [37–39]. This demonstrates that the longitudinal component, although small, remains finite even within the paraxial regime.

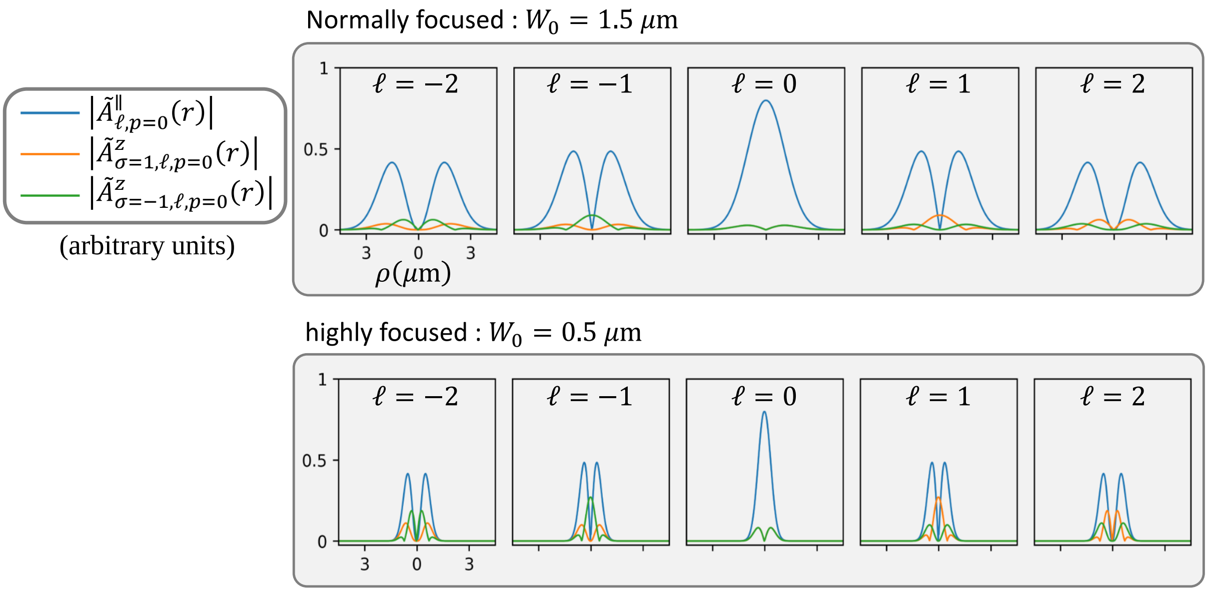

Figure 2.5 : Magnitude (absolute value) of $\boldsymbol{A}_{\sigma,\ell,p=0}(\boldsymbol{r})$ at $z = 0$ for different components and spin. The $\sigma\ell$ term leads to asymmetry between cases with the same ℓ in the longitudinal component (a wavelength of 532 nm is assumed).

As for the $\sigma + \ell$ dependence in the phase term, it is retained in the angular spectrum representation and will be discussed below. By combining Eq. (2.23) with the two-dimensional Fourier transformation defined in Coulomb gauge, vector potential can be written as

$$\boldsymbol{A}_{\sigma,\ell,p}(\boldsymbol{r}) = \frac{1}{(2\pi)^2} \int \mathrm{d}^2 \boldsymbol{q}_\parallel \, \boldsymbol{\mathcal{A}}_{\sigma,\ell,p}(\boldsymbol{q}_\parallel) e^{i\boldsymbol{q}_\parallel \cdot \boldsymbol{r}_\parallel} e^{iq_0 z}. \tag{2.26}$$

Since we primarily focus on the fundamental radial mode $p = 0$, from now on we omit the subscript p to simplify the notations. The complex amplitude under Coulomb gauge in the angular spectrum representation is then written as

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell}(\boldsymbol{q}_\parallel) = \boldsymbol{\mathcal{A}}_{\sigma,\ell,p=0}(\boldsymbol{q}_\parallel) = \tilde{A}^{\parallel}_{\ell}(\boldsymbol{q}_\parallel)\hat{\boldsymbol{\varepsilon}}_\sigma + \tilde{A}^{z}_{\sigma,\ell}(\boldsymbol{q}_\parallel)\hat{\boldsymbol{z}}. \tag{2.27}$$

The corresponding complex amplitude in the angular spectrum representation is obtained as

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell,p}(\boldsymbol{q}_\parallel) = \left[\hat{\boldsymbol{\varepsilon}}_\sigma - \left(\frac{|\boldsymbol{q}|}{q_0}\right)^2 (\hat{\boldsymbol{\varepsilon}}_\sigma \cdot \hat{\boldsymbol{q}}_\parallel)\hat{\boldsymbol{q}}\right] \mathcal{A}^L_{\ell,p}(\boldsymbol{q}_\parallel) \tag{2.28}$$

$$= \left[\hat{\boldsymbol{\varepsilon}}_\sigma - \left(\frac{|\boldsymbol{q}|}{q_0}\right)^2 \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} e^{i\sigma\phi_{\boldsymbol{q}_\parallel}} \hat{\boldsymbol{q}}\right] \mathcal{A}^L_{\ell,p}(\boldsymbol{q}_\parallel) \tag{2.29}$$

$$\approx \left[\hat{\boldsymbol{\varepsilon}}_\sigma - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} e^{i\sigma\phi_{\boldsymbol{q}_\parallel}} \hat{\boldsymbol{z}}\right] \mathcal{A}^L_{\ell,p}(\boldsymbol{q}_\parallel), \tag{2.30}$$

where the final approximation follows from the paraxial condition $\cos\theta_{\boldsymbol{q}} = q_0/|\boldsymbol{q}| \approx 1$. The wavevector unit vector is written as $\hat{\boldsymbol{q}} = \sin\theta_{\boldsymbol{q}}\left(\cos\phi_{\boldsymbol{q}_\parallel}\hat{\boldsymbol{x}} + \sin\phi_{\boldsymbol{q}_\parallel}\hat{\boldsymbol{y}}\right) + \cos\theta_{\boldsymbol{q}}\hat{\boldsymbol{z}} \approx \hat{\boldsymbol{z}}$, and $\sin\theta_{\boldsymbol{q}} = \frac{q_\parallel}{\sqrt{q_\parallel^2+q_0^2}}$. Substituting the complex amplitude under Lorenz gauge derived previously, each component of the complex amplitude takes the form

$$\begin{aligned} \tilde{\mathcal{A}}^\parallel_\ell(\boldsymbol{q}_\parallel) &\equiv \tilde{\mathcal{A}}^\parallel_{\ell,p=0}(\boldsymbol{q}_\parallel) \approx \tilde{F}_{|\ell|}(q_\parallel)e^{i\ell\phi_{\boldsymbol{q}_\parallel}} \\ \tilde{\mathcal{A}}^z_{\sigma,\ell}(\boldsymbol{q}_\parallel) &\equiv \tilde{\mathcal{A}}^z_{\sigma,\ell,p=0}(\boldsymbol{q}_\parallel) \approx -\frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{|\ell|}(q_\parallel)e^{iJ\phi_{\boldsymbol{q}_\parallel}}, \end{aligned} \tag{2.31}$$

where $\tilde{F}_{|\ell|}(q_\parallel) = \tilde{F}_{|\ell|,p=0}(q_\parallel)$. This expression shows that the SOI is encoded in the phase of the longitudinal component through the combined dependence $(\sigma + \ell)$, corresponding to the total angular momentum (TAM). This behavior difference is illustrated in Figure 2.6 and Figure 2.7. However, this phase structure is generally difficult to detext experimentally, as it is averaged out in intensity-based measurements. To access this TAM-dependent phase, interference between two LG beams can be employed, forming vector vortex beams (VVBs), which will be discussed in the following section.

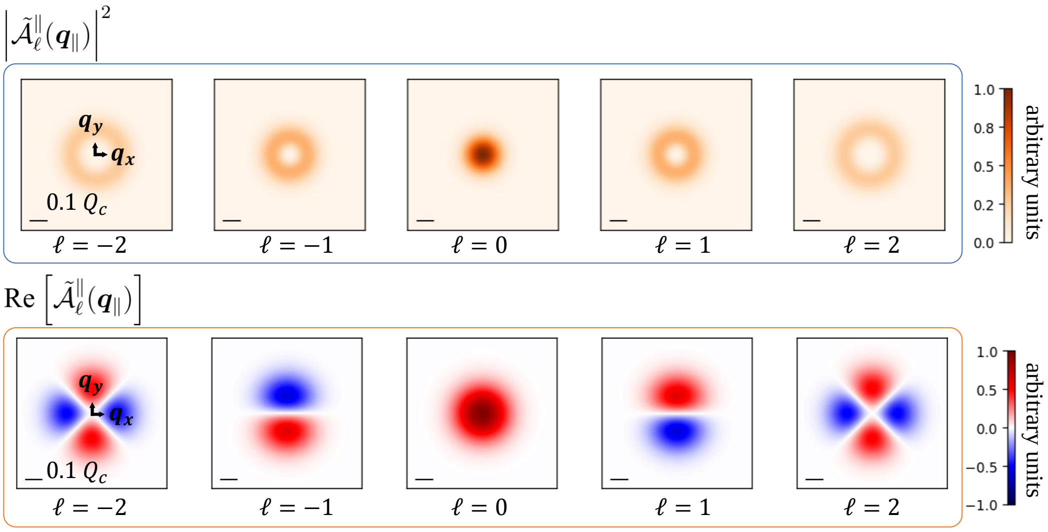

Figure 2.6：Squared magnitude (intensity) and real part distribution of $\tilde{\mathcal{A}}^\parallel_\ell(\boldsymbol{q}_\parallel)$. The scale bar indicates 0.1 times the light-cone radius for WSe_2 monolayers, $Q_c = 8.6\ \mu\text{m}^{-1}$.

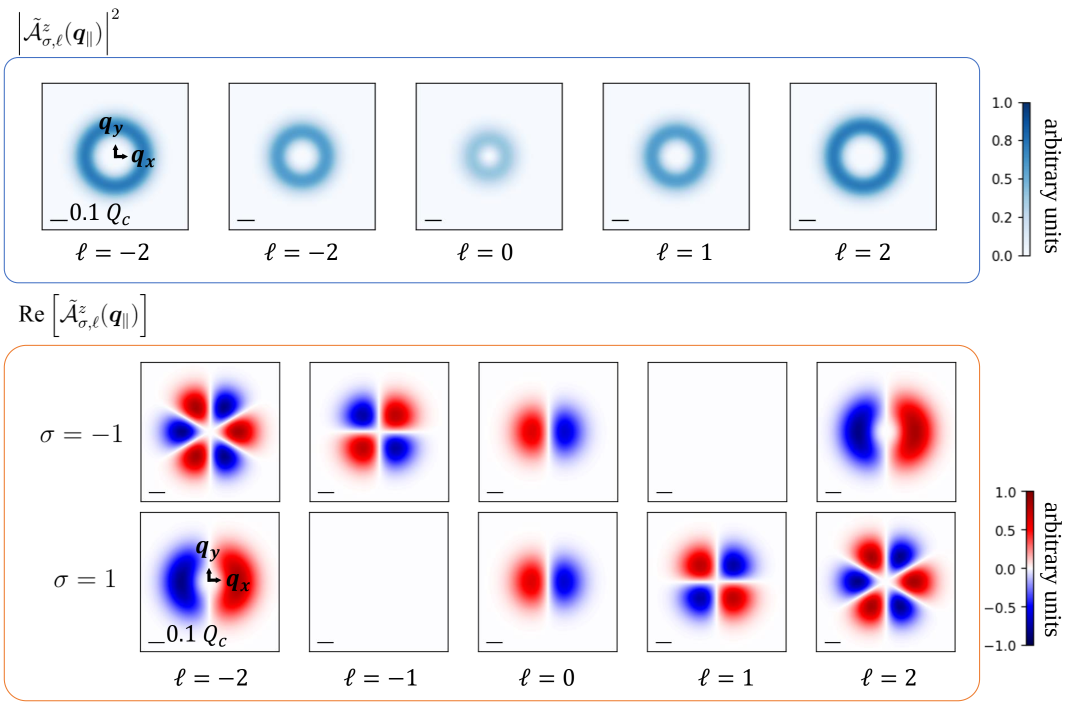

Figure 2.7：Squared magnitude (intensity) and real part distribution of $\tilde{A}^z_{\sigma,\ell}(\boldsymbol{q}_\parallel)$. The scale bar indicates 0.1 times the light-cone radius for WSe_2 monolayers, $Q_c = 8.6\ \mu\mathrm{m}^{-1}$.

2.2.3 Longitudinal Component in Vector Vortex Beam (VVB)

Subject to decoherence, the information of TAM encoded in the phase is likely lost when calculating optical transition rates, as pointed out by Oscar Javier Gomez Sanchez *et al.* in Ref. [30]. To retain and control this information, they introduced vector vortex beams (VVBs), which are constructed by superimposing two twisted light beams , which in general, these beams carry angular momentaums of the same magnitude but opposite signs. Thus, VVB carries vetoral polarization that varies across the transverse plane. In contrast, scalar vortex beam, single twisted light beams, carries uniform polarization. The VVB state is expressed as a superposition through

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta) = \cos\left(\frac{\beta}{2}\right)\boldsymbol{\mathcal{A}}_{\sigma,\ell}(\boldsymbol{q}_\parallel) + e^{i\alpha}\sin\left(\frac{\beta}{2}\right)\boldsymbol{\mathcal{A}}_{\sigma',\ell'}(\boldsymbol{q}_\parallel). \tag{2.32}$$

The parameter α controls the relative phase between the two beams, while β determines their relative weight in the superposition. Throughout this thesis, the radial mode index is set to $p = 0$, as the primary focus remains on the interplay between SAM and OAM. σ' and ℓ' denote the second LG beam that carrying a different combination of angular momentum .

For the folowing disscussion, we define the difference and average of OAM/SAM to enphasize their contribution as

$$\Delta\ell \equiv \ell' - \ell,\ \bar{\ell} \equiv \frac{\ell' + \ell}{2},\ \Delta\sigma \equiv \sigma' - \sigma,\ \bar{\sigma} \equiv \frac{\sigma' + \sigma}{2}.$$

A VVB state also contains both transverse and longitudinal components, which can be written as

$$\boldsymbol{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta) = \tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta) + \tilde{\boldsymbol{\mathcal{A}}}^{z}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta). \tag{2.33}$$

where the polarization in the transverse component is no longer a fixed unit vector, but becomes $\boldsymbol{q}_\parallel$ dependent as a consequence of the superposition. Therefore, it cannot be factored out as a constant vector in the angular spectrum.

Transverse Components of VVBs :

The transverse component is given by

$$\begin{aligned}
&\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta) = \tilde{\mathcal{A}}^{x}_{\ell,\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta)\hat{\boldsymbol{x}} + \tilde{\mathcal{A}}^{y}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta)\hat{\boldsymbol{y}} \\
&= \frac{1}{\sqrt{2}} e^{i\left(\bar{\ell}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)} \Bigg\{ \cos\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell|}(q_\parallel) + \sin\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell'|}(q_\parallel)\right] \\
&\qquad - i\sin\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell|}(q_\parallel) - \sin\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell'|}(q_\parallel)\right]\Bigg\}\hat{\boldsymbol{x}} \\
&+ \frac{i\bar{\sigma}}{\sqrt{2}} e^{i\left(\bar{\ell}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)} \Bigg\{ \cos\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell|}(q_\parallel) + \sin\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell'|}(q_\parallel)\right] \\
&\qquad - i\sin\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell|}(q_\parallel) - \sin\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell'|}(q_\parallel)\right]\Bigg\}\hat{\boldsymbol{y}} \\
&- \frac{\Delta\sigma}{2}\frac{1}{\sqrt{2}} e^{i\left(\bar{\ell}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)} \Bigg\{ \sin\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell|}(q_\parallel) + \sin\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell'|}(q_\parallel)\right] \\
&\qquad + i\cos\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)\left[\cos\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell|}(q_\parallel) - \sin\left(\frac{\beta}{2}\right)\tilde{F}_{|\ell'|}(q_\parallel)\right]\Bigg\}\hat{\boldsymbol{y}}
\end{aligned} \tag{2.34}$$

Since Eq. (2.34) is too complex, we adpot the condition below to focus on the cases that we might be interested.

$$\sigma = -\sigma' = 1, \beta = \pi/2$$

With the definition

$$\begin{aligned}
\text{sum of two radial profiles :}\quad & G_{|\ell|,|\ell'|}(q_\parallel) \equiv F_{|\ell|}(q_\parallel) + F_{|\ell'|}(q_\parallel), \\
\text{difference of two radial profiles :}\quad & H_{|\ell|,|\ell'|}(q_\parallel) \equiv F_{|\ell|}(q_\parallel) - F_{|\ell'|}(q_\parallel)
\end{aligned}$$

the formulation can then reduce to

$$\begin{aligned}
&\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta=\pi/2) = \frac{1}{2}e^{i\bar{\ell}\phi_{\boldsymbol{q}_\parallel}}\, e^{i\left(-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)} \\
&\times\Bigg\{\left[\cos\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}-\frac{\pi}{2}\Delta|\ell|+\frac{\alpha}{2}\right)G_{|\ell|,|\ell'|}(q_\parallel) - i\sin\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}-\frac{\pi}{2}\Delta|\ell|+\frac{\alpha}{2}\right)H_{|\ell|,|\ell'|}(q_\parallel)\right]\hat{\boldsymbol{x}} \\
&\quad+\left[\sin\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}-\frac{\pi}{2}\Delta|\ell|+\frac{\alpha}{2}\right)G_{|\ell|,|\ell'|}(q_\parallel) + i\cos\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}-\frac{\pi}{2}\Delta|\ell|+\frac{\alpha}{2}\right)H_{|\ell|,|\ell'|}(q_\parallel)\right]\hat{\boldsymbol{y}}\Bigg\},
\end{aligned} \tag{2.35}$$

where $\Delta|\ell| \equiv |\ell'| - |\ell|$, $\overline{|\ell|} \equiv \frac{|\ell'|+|\ell|}{2}$, originate from the radial profile, $\tilde{F}_{|\ell|}(q_\parallel) = (-i)^{|\ell|}F_{|\ell|}(q_\parallel)$.

This result shows that the amplitude and polarization can be separated by imposing

$$|\ell| = |\ell'|,$$

under which $G_{|\ell|,|-\ell|}(q_\parallel) = 2F_{|\ell|}(q_\parallel)$ while $H_{|\ell|,|-\ell|}(q_\parallel) = 0$. The transverse conponent can then be written as

$$\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta=\pi/2) \equiv \tilde{\mathcal{A}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel)\,\hat{\boldsymbol{\varepsilon}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel;\alpha), \tag{2.36}$$

where the amplitude and polarization are given by

$$\begin{aligned} \tilde{\mathcal{A}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel) &= \tilde{F}_{|\ell|}(q_\parallel)\, e^{i\bar{\ell}\phi_{\boldsymbol{q}_\parallel}}, \\ \hat{\boldsymbol{\varepsilon}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel;\alpha) &= e^{i\alpha/2}\left[\cos\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)\hat{\boldsymbol{x}} + \sin\left(\frac{\Delta\ell}{2}\phi_{\boldsymbol{q}_\parallel}+\frac{\alpha}{2}\right)\hat{\boldsymbol{y}}\right]. \end{aligned} \tag{2.37}$$

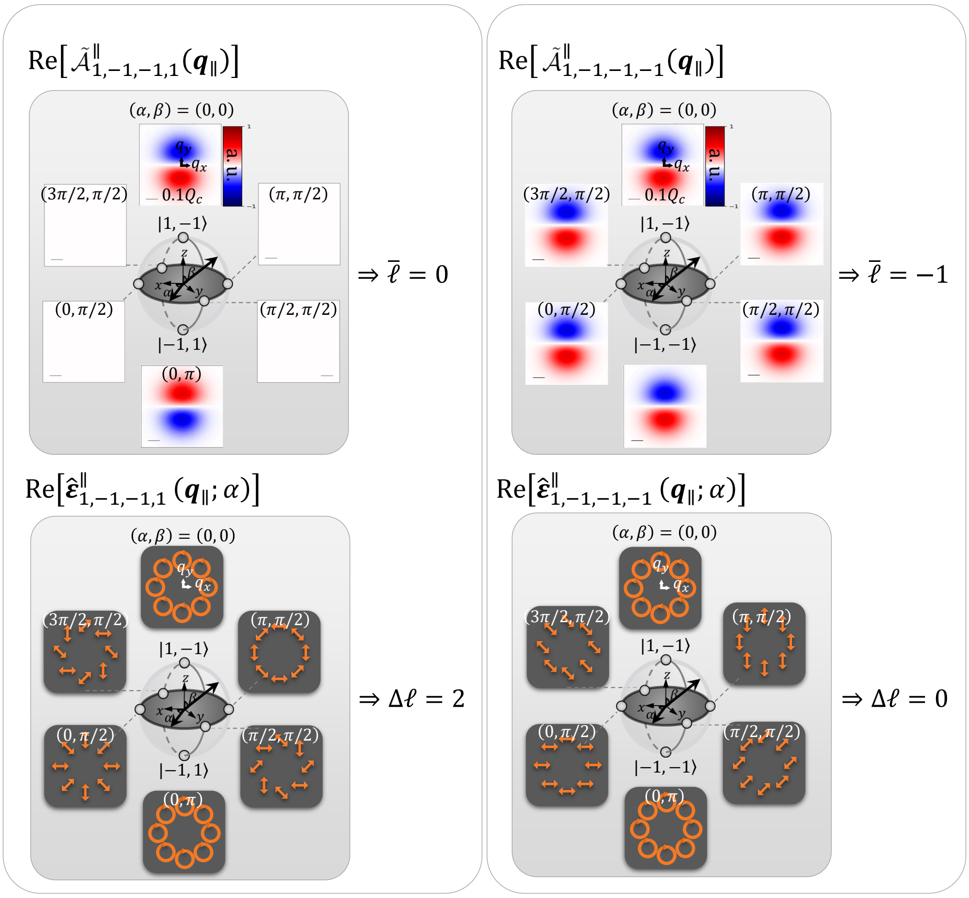

Figure 2.8 : Real part and the polarization of $\tilde{\boldsymbol{\mathcal{A}}}^{\parallel}_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel;\alpha,\beta=\pi/2)$. The corresponding VVB states are represented on the Poincaré sphere, where the north and south poles correspond to the basis states $|\sigma,\ell\rangle$ and $|\sigma',\ell'\rangle$, respectively.

These expressions show that, on the equator of the Poincaré sphere ($\beta = \pi/2$), the amplitude of the transverse component retains the same radial profile as the constituent LG beams, while its phase carries the average OAM, $\bar{\ell}$. Meanwhile, the polarization is purely linear, with its orientation determined by both the initial phase α and the OAM difference $\Delta\ell$. Depending on whether the constituent beams carry the same or opposite OAM ($\ell' = \pm\ell$), the OAM difference becomes

$$\Delta\ell = \begin{cases} 0, & \text{for } \ell' = \ell, \\ -2\ell, & \text{for } \ell' = -\ell, \end{cases}$$

respectively. Consequently, the polarization acquires an azimuthal dependence, giving rise to the characteristic spatially varying polarization of a vector vortex beam, which contrasts with the uniform polarization of a scalar vortex beam (i.e., a single LG beam). Figure 2.8 shows how $\bar{\ell}$ and $\Delta\ell$ are related to the phase and polarization of the transverse conponent.

Longitudinal Components of VVBs :

With the definition of difference and average of TAM between the two constituent beams

$$\Delta J \equiv J' - J,\ \bar{J} \equiv \frac{J' + J}{2},$$

the longitudinal component is given in the same condition $\left(\sigma = -\sigma',\ \beta = \frac{\pi}{2},\ \text{and } |\ell| = |\ell'|\right)$ by

$$\tilde{\mathcal{A}}^z_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel; \alpha, \beta = \pi/2) \equiv \tilde{A}^z_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel)\, \hat{\boldsymbol{\varepsilon}}^z_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel; \alpha), \tag{2.38}$$

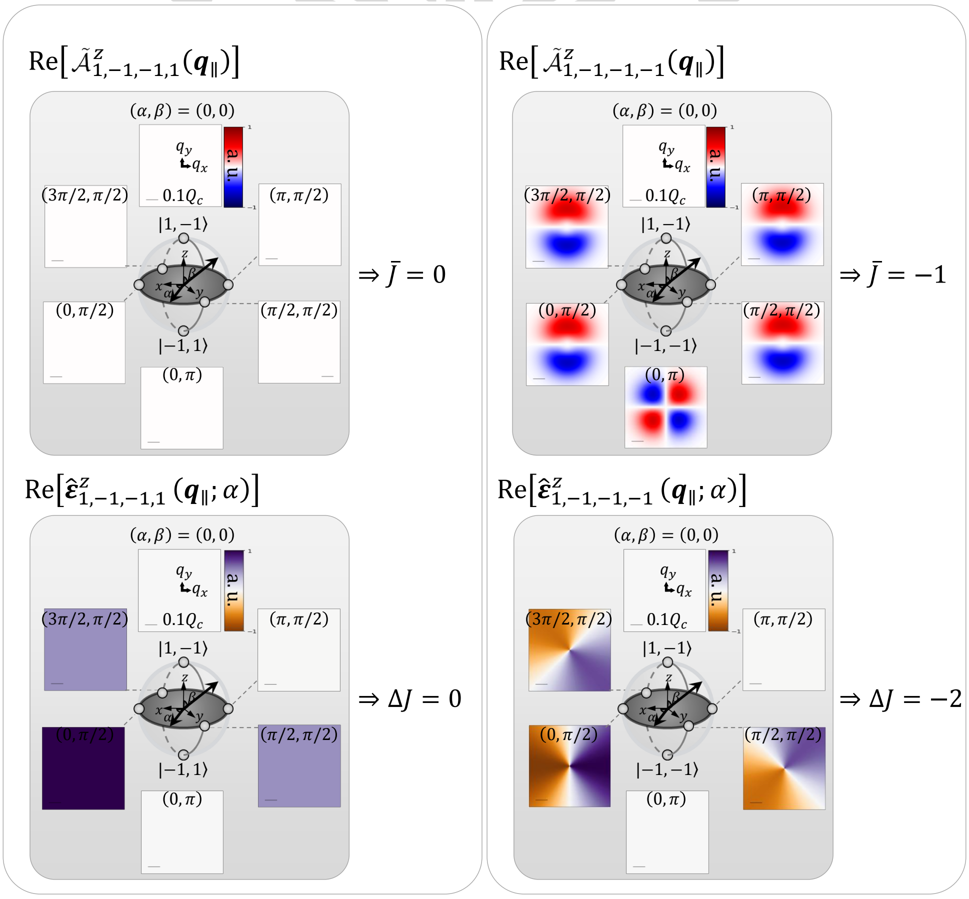

Figure 2.9 : Real part and the polarization of $\tilde{\mathcal{A}}^z_{1,\ell,-1,\ell'}(\boldsymbol{q}_\parallel; \alpha, \beta = \pi/2)$. The corresponding VVB states are represented on the Poincaré sphere, where the north and south poles correspond to the basis states $|\sigma, \ell\rangle$ and $|\sigma', \ell'\rangle$, respectively

and the corresponding amplitude and polarization are given by

$$\begin{aligned}\tilde{\mathcal{A}}^{z}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel}) &= -\frac{\sin\theta_{\boldsymbol{q}}}{2}\tilde{F}_{|\ell|}(q_{\parallel})\, e^{i\bar{J}\phi_{\boldsymbol{q}_{\parallel}}}, \\ \hat{\boldsymbol{\varepsilon}}^{z}_{1,\ell,-1,\ell'}(\boldsymbol{q}_{\parallel};\alpha) &= e^{i\alpha/2}\cos\left(\frac{\Delta J}{2}\phi_{\boldsymbol{q}_{\parallel}} + \frac{\alpha}{2}\right)\hat{\boldsymbol{z}}.\end{aligned} \tag{2.39}$$

Unlike the transverse component, the phase of the longitudinal component is characterized by the average TAM, $\bar{J}$, while its polarization is governed by both the initial phase α and the TAM difference ΔJ. Depending on whether the constituent beams carry the same or opposite OAM ($\ell' = \pm\ell$), the TAM difference becomes

$$\Delta J = \begin{cases} -2, & \text{for } \ell' = \ell, \\ -2\ell - 2, & \text{for } \ell' = -\ell, \end{cases}$$

respectively. Consequently, the longitudinal polarization exhibits an azimuthal dependence determined by ΔJ, which, similar to the transverse component, is a distinctive feature of vector vortex beams and is absent in scalar vortex beams. Figure 2.9 illustrates how the average TAM, $\bar{J}$, and the TAM difference, ΔJ, are related to the phase and polarization of the longitudinal conponent.

Relaxed condition —— $|\ell| \approx |\ell'|$:
When the condition $|\ell| = |\ell'|$ is relaxed, the transverse component reduces to Eq. (2.35), while the longitudinal component becomes

$$\begin{aligned}&\boldsymbol{\mathcal{A}}^{z}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta=\pi/2) = -\frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}} e^{i\bar{J}\phi_{\boldsymbol{q}_{\parallel}}} e^{i\left(-\frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2}\right)} \\ &\times\left\{\cos\left(\frac{\Delta J}{2}\phi_{\boldsymbol{q}_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) G_{|\ell|,|\ell'|}(q_{\parallel}) - i\sin\left(\frac{\Delta J}{2}\phi_{\boldsymbol{q}_{\parallel}} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right) H_{|\ell|,|\ell'|}(q_{\parallel})\right\}\hat{\boldsymbol{z}}.\end{aligned} \tag{2.40}$$

Although the amplitude and polarization can no longer be separated in the same manner as in the case of $|\ell| = |\ell'|$, we can still be analyze it by considering the approximate condition

$$|\ell| \approx |\ell'|,$$

for which

$$G_{|\ell|,|\ell'|}(q_{\parallel}) \gg H_{|\ell|,|\ell'|}(q_{\parallel}).$$

Under this approximation, the dominant contribution is governed by the first term, which retains the same essential features discussed previously. In particular, the longitudinal conponent possesses a helical phase, $e^{i\bar{J}\phi_{\boldsymbol{q}_{\parallel}}}$, characterized by the average TAM, $\bar{J}$ and a modulation primarily determined by the TAM difference, ΔJ. Interestingly, relaxing the condition $|\ell| = |\ell'|$ also relaxes the allowed values of the TAM difference, which now becomes

$$\Delta J = \ell' - \ell - 2.$$

Consequently, ΔJ is no longer restricted to even integers, allowing odd values of ΔJ. This introduces new types of distributions, for example, the longitudinal conponent $\boldsymbol{\mathcal{A}}^z_{1,-1,-1,2}(\boldsymbol{q}_\parallel; \alpha, \beta = \pi/2)$ corresponds to $\Delta J = 1$, as shown in Fig. 2.10.

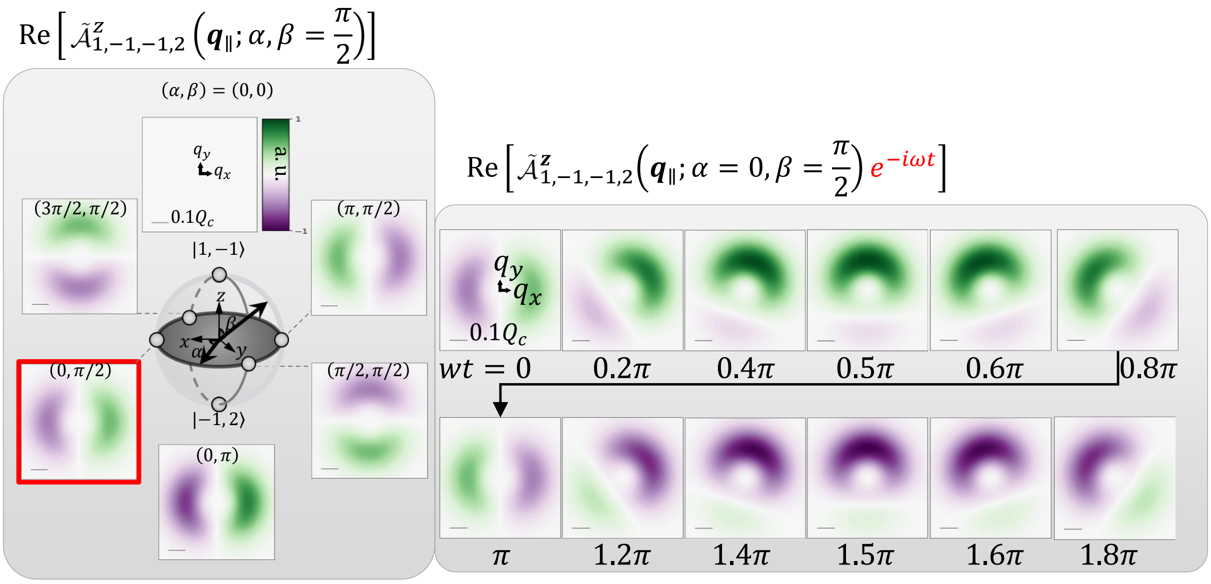

Figure 2.10 : Real part of $\boldsymbol{\mathcal{A}}^z_{1,-1,-1,2}(\boldsymbol{q}_\parallel; \alpha, \beta = \pi/2)$ and its time evolution for the case $(\alpha = 0, \beta = \pi/2)$.

Mathematically, the dominant contribution vanishes at

$$\phi_{\boldsymbol{q}_\parallel} = \frac{\pi}{2} - \alpha \quad \text{or} \quad \phi_{\boldsymbol{q}_\parallel} = \frac{3\pi}{2} - \alpha,$$

thereby giving rise to the two petal structure as shown in the left hand side of Fig. 2.10. However, from a physical perspective, the helical phase characterized by $\bar{J}$ should be considered separately. After factoring out this helical phase, the remaining angular modulation is

$$-e^{i\left(-\frac{\pi}{2}\overline{|\ell|}+\frac{\alpha}{2}\right)} \cos\left(\frac{\Delta J}{2}\phi_{\boldsymbol{q}_\parallel} - \frac{\pi}{4}\Delta|\ell| + \frac{\alpha}{2}\right),$$

which, for the case $\boldsymbol{\mathcal{A}}^z_{1,-1,-1,2}(\boldsymbol{q}_\parallel; \alpha, \beta = \pi/2)$, whose real part reaches its extrema at

$$\phi_{\boldsymbol{q}_\parallel} = \frac{\pi}{2} - \alpha.$$

This indicates that during the time evolution (right panel of Fig. 2.10), the helical phase associated with $\bar{J}$ rotates continuously relative to the modulation from ΔJ. Consequently, the distribution changes with the orientation of the petals. When the petals are aligned with the direction corresponding to $\phi_{\boldsymbol{q}_\parallel} = \frac{\pi}{2} - \alpha$, the maximum contribution is obtained, leading to an asymmetric patterns. This implies that, when $\Delta J = 1$, one can generate a light field whose dominant plane-wave component is localized around a preferred direction in the $\boldsymbol{q}_\parallel$ plane. Consequently, such a field may provide a means to selectively excite gray excitons with a preferred center-of-mass momentum, as proposed by Oscar Javier Gomez Sanchez *et al.* in Ref. [30].

Chapter 3 Light-Matter Interaction

3.1 Theory of Light–Matter Interaction

3.1.1 Fermi's golden rule

In this section, we examine the interaction between light and matter using **Fermi's golden rule**. The transition rate for an optical excitation from the ground state $|GS\rangle$, with fully occupied valence bands, to an exciton state $|\Psi^{S,\boldsymbol{Q}}\rangle$ with band index S and center-of-mass momentum $\boldsymbol{Q}$ is given by

$$\Gamma = \frac{2\pi}{\hbar}\left|\langle\Psi^{S,\boldsymbol{Q}}|\hat{H}_I|GS\rangle\right|^2 \delta(E^{S,\boldsymbol{Q}} - \hbar\omega). \tag{3.1}$$

The delta function enforces energy conservation between the exciton energy $E^{S,\boldsymbol{Q}}$ and the photon energy $\hbar\omega$. Within the pseudospin model [6, 30, 40], the exciton state is written as

$$|\Psi^{S,\boldsymbol{Q}}\rangle = \frac{1}{\sqrt{\Omega}}\sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k})\hat{c}^{\dagger}_{c,\boldsymbol{k}+\boldsymbol{Q}}\hat{h}^{\dagger}_{v,-\boldsymbol{k}}|GS\rangle. \tag{3.2}$$

Here, the particle operators $\hat{c}^{\dagger}_{c,\boldsymbol{k}}$ and $\hat{h}^{\dagger}_{v,-\boldsymbol{k}}$ create an electron in the conduction band with crystal momentum $\boldsymbol{k}$ and a hole in the valence band with crystal momentum $-\boldsymbol{k}$, respectively. The coefficient $\Lambda_{S,\boldsymbol{Q}}(vc\boldsymbol{k})$ represents the amplitude of the electron–hole configuration for exciton band S, and Ω is the area of the TMD monolayer. The matrix element in Eq. (3.1) can be expressed, within the **rotating wave approximation** [41], as

$$\langle\Psi^{S,\boldsymbol{Q}}|\hat{H}_I|GS\rangle = \frac{1}{\sqrt{\Omega}}\sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}*}(vc\boldsymbol{k})\langle\psi_{c,\boldsymbol{k}+\boldsymbol{Q}}|H_I|\psi_{v,\boldsymbol{k}}\rangle, \tag{3.3}$$

where, for weak fields, the time-independent part of light–matter interaction Hamiltonian is given by [6, 30]

$$H_I = \frac{|e|}{2m_e}\boldsymbol{A}(\boldsymbol{r})\cdot\boldsymbol{p},$$

with $\boldsymbol{A}(\boldsymbol{r})$ the vector potential in the **Coulomb gauge**, $\boldsymbol{p}$ the momentum operator, and m_e (e) the electron mass (charge). If the vector potential is expanded in a **plane-wave basis** as in Eq. (2.26), the matrix element in Eq. (3.3) becomes

$$\begin{aligned}\langle\psi_{c,\boldsymbol{k}+\boldsymbol{Q}}|H_I|\psi_{v,\boldsymbol{k}}\rangle &= \frac{|e|}{2m_e}\sum_{\boldsymbol{q}_\parallel}\left[\boldsymbol{\mathcal{A}}(\boldsymbol{q}_\parallel)\cdot\langle\psi_{c,\boldsymbol{k}+\boldsymbol{Q}}|e^{i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}}\hat{\boldsymbol{p}}|\psi_{v,\boldsymbol{k}}\rangle\right]\\ &\approx \frac{|e|}{2m_e}\sum_{\boldsymbol{q}_\parallel}\left[\boldsymbol{\mathcal{A}}(\boldsymbol{q}_\parallel)\cdot\delta_{\boldsymbol{q}_\parallel,\boldsymbol{Q}}\langle\psi_{c,\boldsymbol{k}}|\hat{\boldsymbol{p}}|\psi_{v,\boldsymbol{k}}\rangle\right],\end{aligned} \tag{3.4}$$

where the approximation follows from the **electric dipole approximation**. In this approximation, the Bloch state wave functions are written as $\psi_{n,\boldsymbol{k}}(\boldsymbol{r}) = u_{n,\boldsymbol{k}}(\boldsymbol{r})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, and the plane-wave components combine to yield the Kronecker delta $\delta_{\boldsymbol{q}_{\parallel},\boldsymbol{Q}}$, enforcing the equality between the in-plane wavevector and the exciton center-of-mass momentum.

Substituting this result back, the matrix element in Eq. (3.3) [6, 30] becomes

$$\langle\Psi^{S,\boldsymbol{Q}}|H_I|GS\rangle = \frac{E_g}{2i\hbar}\left(\boldsymbol{\mathcal{A}}(\boldsymbol{Q})\cdot\boldsymbol{D}^{S,\boldsymbol{Q}*}\right), \tag{3.5}$$

where superscript * denotes complex conjugation, E_g is the band gap at the K/K' valleys and $\boldsymbol{D}^{S,\boldsymbol{Q}}$ is the exciton transition dipole moment , which is defined as

$$\boldsymbol{D}^{S,\boldsymbol{Q}} = \frac{e\hbar}{im_0E_g\sqrt{\Omega}}\sum_{vc\boldsymbol{k}}\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k})\langle\psi_{v,\boldsymbol{k}}|\boldsymbol{p}|\psi_{c,\boldsymbol{k}}\rangle \tag{3.6}$$

The transition dipole moment obtained from density functional theory combined with the Bethe–Salpeter equation, is given in the small-$\boldsymbol{Q}$ limit by Ref [6, 30]:

Table 3.1：Table of transition dipole moments with corresponding exciton types.

Exciton Type	**Transition Dipole Moment**
Valley excitons (bright)	$\boldsymbol{D}^{K,\boldsymbol{Q}} = \frac{D^B}{\sqrt{2}}(\hat{\boldsymbol{x}}+i\hat{\boldsymbol{y}})$, $\boldsymbol{D}^{K',\boldsymbol{Q}} = \frac{D^B}{\sqrt{2}}(\hat{\boldsymbol{x}}-i\hat{\boldsymbol{y}})$
Valley-mixed excitons (bright)	$\boldsymbol{D}^{B+,\boldsymbol{Q}} = D^B\left(\cos\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{x}}+\sin\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{y}}\right)$, $\boldsymbol{D}^{B-,\boldsymbol{Q}} = iD^B\left(-\sin\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{x}}+\cos\phi_{\boldsymbol{Q}}\,\hat{\boldsymbol{y}}\right)$
Gray excitons	$\boldsymbol{D}_{G,\boldsymbol{Q}} = D^G\,\hat{\boldsymbol{z}}$.

Here, the subscript symbols K, K', $B\pm$, and G denote the K valley, K' valley, upper/ lower band, and gray exciton, respectively. The magnitude of the dipole moment of BXs (GX) is D^B (D^G) [6, 30].

3.2 Manifestation of SOI in Excitonic Optical Transitions

3.2.1 Optical Transition Pattern of Excitons

Under the excitaion of VVBs the corresponding Fermi's golden rule can then be written as [30]

$$\Gamma^{S,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}} = \frac{2\pi}{\hbar}\left|\tilde{M}^{S,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}\right|^2\rho(\hbar\omega = E^{S,\boldsymbol{Q}}), \tag{3.7}$$

where the optical matrix element is defined as

$$\tilde{M}^{S,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}} \equiv \frac{1}{\sqrt{\Omega}} \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}*}(vc\boldsymbol{k}) \left\langle \psi_{c,\boldsymbol{k}+\boldsymbol{Q}} \left| \frac{|e|}{2m_e} \boldsymbol{\mathcal{A}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{Q};\alpha,\beta) \cdot \boldsymbol{p} \right| \psi_{v,\boldsymbol{k}} \right\rangle$$
$$\approx \frac{E_g}{2i\hbar} \left(\mathcal{A}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{Q};\alpha,\beta) \cdot \boldsymbol{D}^{S,\boldsymbol{Q}*} \right). \tag{3.8}$$

The approximation follows from the electric dipole approximation. This expression shows that optical excitation is governed by the "alignment between the light polarization and the transition dipole moment". The transition rate is therefore proportional to both the magnitude of the transition dipole moment and the magnitude of the vector potential. With Eq. (**??**), we can first derive the optical matrix elememts for BXs and GX as

$$\begin{aligned}
\tilde{M}^{B+,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{Q};\alpha,\beta) = &-\frac{E_g D^B}{2\sqrt{2}\hbar} \exp\left\{ i \left[(\bar{\ell}+\bar{\sigma})\phi_{\boldsymbol{Q}} + \alpha \right] \right\} \\
&\times \left\{ \cos\left[\tfrac{1}{2}(\Delta\ell + \Delta\sigma)\phi_{\boldsymbol{Q}} + \frac{\alpha}{2} \right] \left[\cos(\beta/2) \tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(Q) + \sin(\beta/2) \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(Q) \right] \right. \\
&\left. - i \sin\left[\tfrac{1}{2}(\Delta\ell + \Delta\sigma)\phi_{\boldsymbol{Q}} + \frac{\alpha}{2} \right] \left[\cos(\beta/2) \tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(Q) - \sin(\beta/2) \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(Q) \right] \right\}, \\
\tilde{M}^{B-,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{Q};\alpha,\beta) = &-\frac{iE_g D^B}{2\sqrt{2}\hbar} \exp\left\{ i \left[(\bar{\ell}+\bar{\sigma})\phi_{\boldsymbol{Q}} + \alpha \right] \right\} \\
&\times \left\{ i\cos\left[\tfrac{1}{2}(\Delta\ell + \Delta\sigma)\phi_{\boldsymbol{Q}} + \frac{\alpha}{2} \right] \left[\sigma\cos(\beta/2) \tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(Q) + \sigma' \sin(\beta/2) \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(Q) \right] \right. \\
&\left. + \sin\left[\tfrac{1}{2}(\Delta\ell + \Delta\sigma)\phi_{\boldsymbol{Q}} + \frac{\alpha}{2} \right] \left[\sigma\cos(\beta/2) \tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(Q) - \sigma' \sin(\beta/2) \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(Q) \right] \right\}, \\
\tilde{M}^{G,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},\Delta\sigma,\bar{\sigma}}(\boldsymbol{Q};\alpha,\beta) = &-\frac{E_g D^G \sin\theta_{\boldsymbol{Q}}}{2\sqrt{2}\hbar} \exp\left\{ i \left[(\bar{\ell}+\bar{\sigma})\phi_{\boldsymbol{Q}} + \alpha \right] \right\} \\
&\times \left\{ \cos\left[\tfrac{1}{2}(\Delta\ell + \Delta\sigma)\phi_{\boldsymbol{Q}} + \frac{\alpha}{2} \right] \left[\cos(\beta/2) \tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(Q) + \sin(\beta/2) \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(Q) \right] \right. \\
&\left. - i \sin\left[\tfrac{1}{2}(\Delta\ell + \Delta\sigma)\phi_{\boldsymbol{Q}} + \frac{\alpha}{2} \right] \left[\cos(\beta/2) \tilde{F}_{|\bar{\ell}-\frac{\Delta\ell}{2}|}(Q) - \sin(\beta/2) \tilde{F}_{|\bar{\ell}+\frac{\Delta\ell}{2}|}(Q) \right] \right\}.
\end{aligned} \tag{3.9}$$

Besides the gray exciton's optical matrix elememts, that of both valley-mixed BXs are having the information of total angular momentum, which have not been seen in the vector potnetial (transverse component). The reason may be that the $\phi_{\boldsymbol{Q}}$ dependence of valley-mixed BXs transition dipoles is related to the polarization textures through Eq. (3.8) (see Figure 3.1). Unlike the gary exciton's case, which shows the amplitude structure of the light, optical matrix elements of valley-mixed BXs show the polarization texture of the light.

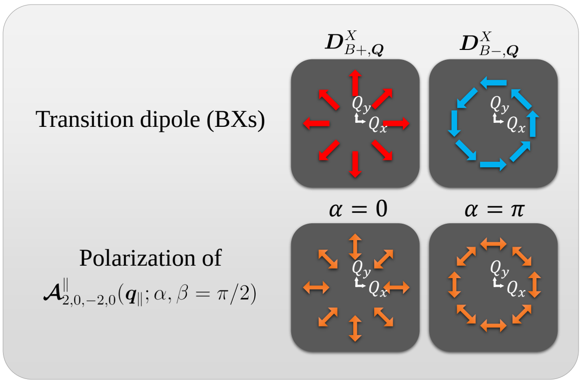

Figure 3.1：Illustration of the transition dipole moments of valley-mixed BXs [6], and their matching with the transverse polarization of the VVB states $\boldsymbol{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_{\parallel};\alpha,\beta) = \boldsymbol{\mathcal{A}}_{1,-1,-1,1}(\boldsymbol{q}_{\parallel};\alpha=0$ and $\alpha=\pi, \beta=\pi/2)$ in momentum space.

Similar to the end of the Chapter 2, we can utilize ΔJ, $\bar{J}$ as well as the defined function, sum/difference of two radial profiles $\left(G_{|\ell|,|\ell'|}(Q)\right.$ and $\left.H_{|\ell|,|\ell'|}(Q)\right)$, to enhpasize the effect of TAM. The optical matrix element with equal weighting and opposite spin

$$\beta = \pi/2\,,\; \Delta\sigma = -2\,,\; \bar{\sigma} = 0$$

is then reduce to

$$\begin{aligned}
\tilde{M}^{B+,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},-2,0}(\boldsymbol{Q};\alpha,\beta=\pi/2) &= \tilde{M}^{B+,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}(\boldsymbol{Q};\alpha,\beta=\pi/2) = -\frac{E_g D^B}{4\hbar}\exp\left\{i\left[\bar{J}\phi_{\boldsymbol{Q}} - \frac{\pi}{2}\overline{|\ell|} + \alpha\right]\right\} \\
&\times\left\{\cos\left[\tfrac{1}{2}\left(\Delta J\phi_{\boldsymbol{Q}} - \frac{\pi}{2}\Delta|\ell| + \alpha\right)\right]G_{|\ell|,|\ell'|}(Q) - i\sin\left[\tfrac{1}{2}\left(\Delta J\phi_{\boldsymbol{Q}} - \frac{\pi}{2}\Delta|\ell| + \alpha\right)\right]H_{|\ell|,|\ell'|}(Q)\right\}, \\
\tilde{M}^{B-,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},-2,0}(\boldsymbol{Q};\alpha,\beta=\pi/2) &= \tilde{M}^{B-,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}(\boldsymbol{Q};\alpha,\beta=\pi/2) = -\frac{iE_g D^B}{4\hbar}\exp\left\{i\left[\bar{J}\phi_{\boldsymbol{Q}} - \frac{\pi}{2}\overline{|\ell|} + \alpha\right]\right\} \\
&\times\left\{\sin\left[\tfrac{1}{2}\left(\Delta J\phi_{\boldsymbol{Q}} - \frac{\pi}{2}\Delta|\ell| + \alpha\right)\right]G_{|\ell|,|\ell'|}(Q) + i\cos\left[\tfrac{1}{2}\left(\Delta J\phi_{\boldsymbol{Q}} - \frac{\pi}{2}\Delta|\ell| + \alpha\right)\right]H_{|\ell|,|\ell'|}(Q)\right\}, \\
\tilde{M}^{G,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},-2,0}(\boldsymbol{Q};\alpha,\beta=\pi/2) &= \tilde{M}^{G,\boldsymbol{Q}}_{\Delta\ell,\bar{\ell},\Delta J,\bar{J}}(\boldsymbol{Q};\alpha,\beta=\pi/2) = -\frac{E_g D^G\sin\theta_{\boldsymbol{Q}}}{4\hbar}\exp\left\{i\left[\bar{J}\phi_{\boldsymbol{Q}} - \frac{\pi}{2}\overline{|\ell|} + \alpha\right]\right\} \\
&\times\left\{\cos\left[\tfrac{1}{2}\left(\Delta J\phi_{\boldsymbol{Q}} - \frac{\pi}{2}\Delta|\ell| + \alpha\right)\right]G_{|\ell|,|\ell'|}(Q) - i\sin\left[\tfrac{1}{2}\left(\Delta J\phi_{\boldsymbol{Q}} - \frac{\pi}{2}\Delta|\ell| + \alpha\right)\right]H_{|\ell|,|\ell'|}(Q)\right\}.
\end{aligned} \tag{3.10}$$

With the condiseration of two beams have similar radial profiles,

$$G_{|\ell|,|\ell'|}(Q) \gg H_{|\ell|,|\ell'|}(Q),$$

The azimuthal dependence of optical matrix elememts for upper band BXs and GX are well approximated by $\cos\left[\frac{1}{2}\left(\Delta J\,\phi_{q_\parallel} - \frac{\pi}{2}\Delta|\ell| + \alpha\right)\right]$. In contrast, that of lower band BXs are well approximated by $\sin\left[\frac{1}{2}\left(\Delta J\,\phi_{q_\parallel} - \frac{\pi}{2}\Delta|\ell| + \alpha\right)\right]$. Thus, it is also possible to create the asymmetrical horseshoe pattern for all the exciton types with the same VVB states (see 123).

If two beams have exactly same radial profiles

$$G_{|\ell|,|\ell'|}(Q) = 2\tilde{F}_{|\ell|}(Q) \text{ and } H_{|\ell|,|\ell'|}(Q) = 0$$

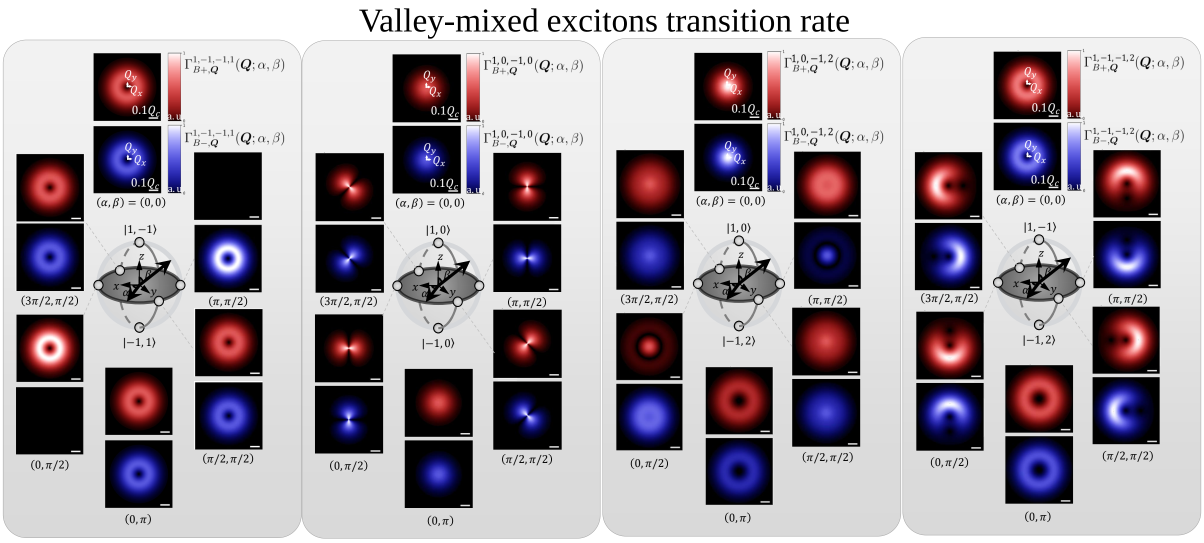

Figure 3.2：Normalized transition rate of valley-mixed BXs excited by the VVB state $|1,-1,-1,1;\alpha,\beta\rangle$ in momentum space. The scale bar indicates the considered 0.1 times light cone range for WSe$_2$-monolayers, $Q_c = 8.6\ \mu\text{m}^{-1}$

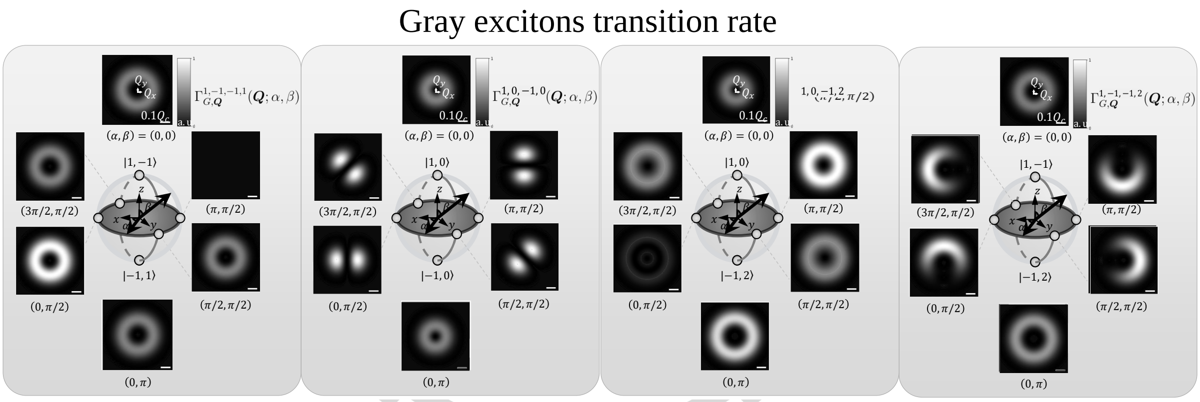

Figure 3.3：Normalized transition rate of GXs excited by the VVB state $|1,-1,-1,1;\alpha,\beta\rangle$ in momentum space.

We now can utilize the formulation above to analyze the SOI in excitonic optical transition for both valley-mixed BXs and GX under VVBs. From Eq. (3.7), the transition rate is propotional to the squared absolute value of optical matrix elememt

$$\Gamma^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'} = \frac{2\pi}{\hbar}\left|\tilde{M}^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\right|^2 \rho(\hbar\omega = E^{S,\boldsymbol{Q}}) \propto \left|\tilde{M}^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\right|^2, \tag{3.11}$$

where

$$\tilde{M}^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'} \propto \left(\boldsymbol{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}(\boldsymbol{Q};\alpha,\beta)\cdot \boldsymbol{D}^{S,\boldsymbol{Q}*}\right). \tag{3.12}$$

This expression has a structure similar to Eq. (2.34) and Eq. (2.40) and can likewise be decomposed into radial, azimuthal, and global phase contributions. How ever, the azimuthal and global phase now all depend on TAM indices...

$$
\begin{aligned}
\Gamma^{B+,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'} &\propto \left|\tilde{M}^{B+,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\right|^2 = \left(\frac{E_g D^B}{2\sqrt{2}\hbar}\right)^2 \left|\cos(\beta/2)\tilde{F}_{\ell,p}(Q) + e^{i(\Delta J \phi_{\boldsymbol{Q}}+\alpha)}\sin(\beta/2)\tilde{F}_{\ell',p}(Q)\right|^2, \\
\Gamma^{B-,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'} &\propto \left|\tilde{M}^{B-,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\right|^2 = \left(\frac{E_g D^B}{2\sqrt{2}\hbar}\right)^2 \left|\sigma\cos(\beta/2)\tilde{F}_{\ell,p}(Q) + e^{i(\Delta J \phi_{\boldsymbol{Q}}+\alpha)}\sigma'\sin(\beta/2)\tilde{F}_{\ell',p}(Q)\right|^2, \\
\Gamma^{G,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'} &\propto \left|\tilde{M}^{G,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\right|^2 = \left(\frac{E_g D^G}{2\sqrt{2}\hbar}\right)^2 \sin^2\theta_{\boldsymbol{Q}} \left|\cos(\beta/2)\tilde{F}_{\ell,p}(Q) + e^{i(\Delta J \phi_{\boldsymbol{Q}}+\alpha)}\sin(\beta/2)\tilde{F}_{\ell',p}(Q)\right|^2
\end{aligned}
\tag{3.13}
$$

In this form, we observe that the SOI term $\Delta J = (\ell' + \sigma') - (\ell + \sigma)$ acts as an additional relative phase modulator with $\phi_{\boldsymbol{Q}}$ dependence. Comparing the cases of valley-mixed BXs and GXs, Eq. (3.5) shows that the spin index σ can enters the phase of the optical matrix element. Consequently, the $\phi_{\boldsymbol{Q}}$ dependence of the transition dipoles for valley-mixed BXs in Eq. (3.1) provides a mechanism for the emergence of ΔJ under excitation by a VVB state, whose polarization texture also depends on $\phi_{\boldsymbol{Q}}$. As a result, valley-mixed BXs can exhibit interference features analogous to those of GXs.

In contrast, for GXs, the SOI manifests directly through the interference term in Eq. (**??**). This originates from the fact that the transition dipole moment of GXs is oriented along the out-of-plane (z) direction and remains isotropic in the in-plane coordinates, making the interference the dominant mechanism responsible for revealing SOI with azimuthal structure. To explicitly reveal the interference effects, we expand the squared matrix elements for valley-mixed BXs as follows:

$$
\begin{aligned}
\left|\tilde{M}^{B+,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\right|^2 &= \left(\frac{E_g D^B}{2\sqrt{2}\hbar}\right)^2 \Bigg\{ \cos^2(\beta/2)\left|\tilde{F}_{\ell,p}(Q)\right|^2 + \sin^2(\beta/2)\left|\tilde{F}_{\ell',p}(Q)\right|^2 \\
&\qquad + \sin\beta\,\mathrm{Re}\left[\tilde{F}_{\ell,p}(Q)\tilde{F}^*_{\ell',p}(Q)e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}\right] \Bigg\}, \\
\left|\tilde{M}^{B-,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\right|^2 &= \left(\frac{E_g D^B}{2\sqrt{2}\hbar}\right)^2 \Bigg\{ \cos^2(\beta/2)\left|\tilde{F}_{\ell,p}(Q)\right|^2 + \sin^2(\beta/2)\left|\tilde{F}_{\ell',p}(Q)\right|^2 \\
&\qquad + \sigma\sigma'\sin\beta\,\mathrm{Re}\left[\tilde{F}_{\ell,p}(Q)\tilde{F}^*_{\ell',p}(Q)e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}\right] \Bigg\}.
\end{aligned}
\tag{3.14}
$$

Similarly, for GXs, we obtain:

$$
\begin{aligned}
\left|\tilde{M}^{G,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\right|^2 = \left(\frac{E_g D_G}{2\sqrt{2}\hbar}\right)^2 \sin^2\theta_{\boldsymbol{Q}} \Bigg\{ &\cos^2(\beta/2)\left|\tilde{F}_{\ell,p}(Q)\right|^2 + \sin^2(\beta/2)\left|\tilde{F}_{\ell',p}(Q)\right|^2 \\
&+ \sin\beta\,\mathrm{Re}\left[\tilde{F}_{\ell,p}(Q)\tilde{F}^*_{\ell',p}(Q)e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}\right] \Bigg\}.
\end{aligned}
\tag{3.15}
$$

In general, the transition rates for BXs and GXs exhibit interference structures analogous to the familiar interference term of the VVB state in Eq. (**??**). In addition, the transition rates of BXs take the same functional form when $\sigma = \sigma'$. Below, we discuss valley-mixed BXs and GX separately, since the physical origins of the SOI term are different. This distinction also affects how the manipulation of the transition rate should be interpreted. Note that one cannot simply set $\sigma = 0$ to represent linearly polarized light in this formulation, because the exponential form has been taken to simplify the derivation of the final bracket in Eq. (3.7).

Valley-mixed Bright Excitons : Polarization-Matching

For valley-mixed BXs, Eq. (3.14) shows that using opposite circular polarizations ($\sigma' = -\sigma$) leads to opposite interference behavior for the two valley-mixed branches. In addition, the SOI term ΔJ, which does not explicitly appear in the transverse intensity profile of the VVB itself, emerges once the phase factor generated by the dot product between the polarization unit vector and the transition dipole moment is included in Eq. (3.5).

Motivated by the polarization texture of the VVB state $|1, -1, -1, 1; \alpha, \beta\rangle$, shown in Figure **??**, the condition $\Delta J = 0$ can be interpreted as a polarization-matching mechanism, illustrated in Figure 3.1. In particular, the polarization of the states

$$|1, -1, -1, 1; \alpha = 0, \beta = \pi/2\rangle \quad \text{and} \quad |1, -1, -1, 1; \alpha = \pi, \beta = \pi/2\rangle$$

matches the directions of the exciton transition dipoles D^B and D^B, respectively (see Figure 3.1). This enables selective excitation of the upper and lower exciton bands with the relative phase α as shown in Figure 3.2.

However, since the valley splitting between valley-mixed BXs is only a few meV and is typically regarded as spectrally unresolvable [42], direct experimental verification of selective excitation may be difficult. Therefore, we consider the transition rate for the total BXs under the condition ($\sigma' = -\sigma$), which is given by

$$\begin{aligned} \Gamma^{B\pm,\boldsymbol{Q}}_{\sigma,\ell,-\sigma,\ell'}(\alpha,\beta) &\propto \sum_{S=B\pm} \left| \tilde{M}^{S,\boldsymbol{Q}}_{\sigma,\ell,-\sigma,\ell'} \right|^2 \\ &= \left(\frac{E_g D^B}{2\hbar} \right)^2 \left[\cos^2\left(\frac{\beta}{2}\right) \left| \tilde{F}_{\ell,p}(Q) \right|^2 + \sin^2\left(\frac{\beta}{2}\right) \left| \tilde{F}_{\ell',p}(Q) \right|^2 \right]. \end{aligned} \tag{3.16}$$

The formulation shows the cancellation of the two interference terms, leading to the disappearance of both the α dependence and the ΔJ dependence. The resulting distribution is shown in Figure 3.4, which exhibits an identical donut-shaped profile over the Poincaré sphere.

Total valley-mixed excitons transition rate

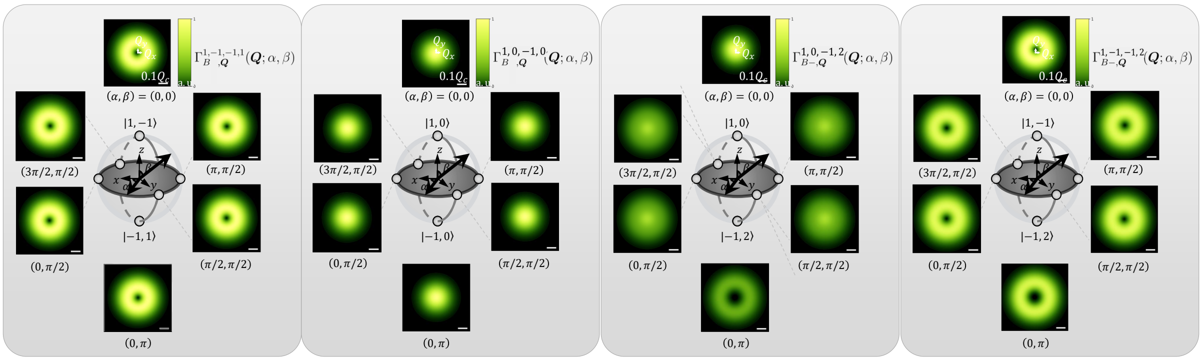

Figure 3.4：Normalized transition rate of total BXs excited by the VVB state $|1,-1,-1,1;\alpha,\beta\rangle$ in momentum space.

Gray Excitons: Manifestation of SOI

For gray excitons, whose transition dipole moments possess only a z component, no additional phase factor is generated by the dot product between the polarization vector and the transition dipole moment in Eq. (3.5). Therefore, the SOI term ΔJ in Eq. (3.15) originates entirely from the longitudinal component of the VVB state in Eq. (**??**). As a result, GX provide an ideal probe of the SOI of light, since their transition rates directly reflect the intrinsic angular-momentum structure of the optical field. At the same time, they offer a possible route for exciting GX using paraxial structured beams.

Further exploit the intensity texture of the VVB state $|1,-1,-1,1;\alpha,\beta\rangle$, shown in Figure **??**. In particular, complete switching of the excitation on and off can be realized by tuning the relative phase between $\alpha=0$ and $\alpha=\pi$ under excitation by the state

$$|1,-1,-1,1;\alpha,\beta=\pi/2\rangle,$$

as shown in Figure 3.5.

Gray excitons transition rate

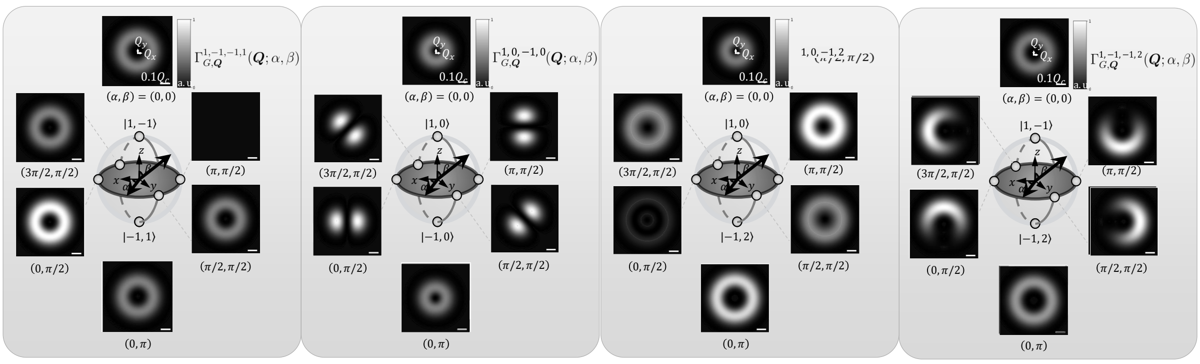

Figure 3.5：Normalized transition rate of GXs excited by the VVB state $|1,-1,-1,1;\alpha,\beta\rangle$ in momentum space.

Beyond complete switching, one may also use the case $\Delta J = 1$ to excite gray excitons with selected finite center-of-mass momentum. For example, consider the state

$$|1, -1, -1, 2; \alpha, \beta = \pi/2\rangle,$$

for which

$$\Delta J = (\ell' + \sigma') - (\ell + \sigma) = 1.$$

First, the radial factor $\tilde{F}_{\ell p}(q_{\|})$ of the two constituent LG beams are no longer identical, so the transition rates at $\beta = 0$ and $\beta = \pi$ (the north and south poles of the Poincaré sphere) is different. Figure **??** shows that the LG beam with larger ℓ produces a larger-radius donut-shaped transition-rate profile in the momentum space.

Second, although the two LG beams do not fully overlap, the condition $\Delta J = 1$ still induces one full interference cycle in the azimuthal direction. Consequently, by tuning α, one can selectively excite GX within a finite range of center-of-mass momentum at $\beta = \pi/2$ (the equator of the Poincaré sphere), as shown in Figure **??**.

3.2.2 Selective Excitation of Excitons

By summing the contributions in momentum space excited by the VVBs, we obtain the total transition rate, which represents the intensity of the material absorption. Under the paraxial approximation, where $\theta_{\boldsymbol{Q}} = \frac{Q}{\sqrt{Q^2+q_0^2}} \approx \frac{Q}{q_0}$ is taken to simplify the integral, the total transition rate can be evaluated as:

$$\Gamma^{S}_{\sigma,\ell,\sigma',\ell'} \equiv \sum_{\boldsymbol{Q}} \Gamma^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'} = \frac{\Omega}{(2\pi)^2} \int_0^\infty \int_0^{2\pi} \Gamma^{S,\boldsymbol{Q}}_{\sigma,\ell,\sigma',\ell'}\, Q \,\mathrm{d}\phi_{\boldsymbol{Q}}\, \mathrm{d}Q.$$

The total transition rates for the upper-band BXs, lower-band BXs, and GXs are obtained as

$$\begin{aligned}
\Gamma^{B+}_{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) &\propto \left|\tilde{M}^{B+}_{\sigma,\ell,\sigma',\ell'}\right|^2 \\
&= \frac{\Phi}{8}\left(\frac{E_g D^B}{a_0 \Omega\hbar}\right)^2 \left\{1+\delta_{\Delta J,0}\sin\beta\cos\left[\frac{\pi}{2}(|\ell'|-|\ell|)+\alpha\right]\frac{\left(\frac{|\ell'|+|\ell|}{2}\right)!}{\sqrt{|\ell'|!|\ell|!}}\right\}, \\
\Gamma^{B-}_{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) &\propto \left|\tilde{M}^{B-}_{\sigma,\ell,\sigma',\ell'}\right|^2 \\
&= \frac{\Phi}{8}\left(\frac{E_g D^B}{a_0 \Omega\hbar}\right)^2 \left\{1+\delta_{\Delta J,0}\sigma\sigma'\sin\beta\cos\left[\frac{\pi}{2}(|\ell'|-|\ell|)+\alpha\right]\frac{\left(\frac{|\ell'|+|\ell|}{2}\right)!}{\sqrt{|\ell'|!|\ell|!}}\right\}, \\
\Gamma^{G}_{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) &\propto \left|\tilde{M}^{G}_{\sigma,\ell,\sigma',\ell'}\right|^2 \\
&= \frac{\Phi}{4}\left(\frac{E_g D_G}{a_0 q_0 W_0 \Omega\hbar}\right)^2 \left\{\left(1+\cos^2\frac{\beta}{2}|\ell|+\sin^2\frac{\beta}{2}|\ell'|\right)\right. \\
&\qquad \left. +\delta_{\Delta J,0}\sin\beta\cos\left[\frac{\pi}{2}(|\ell'|-|\ell|)+\alpha\right]\frac{\left(\frac{|\ell'|+|\ell|}{2}+1\right)!}{\sqrt{|\ell'|!|\ell|!}}\right\},
\end{aligned} \tag{3.17}$$

(YOU FORGET TO ADD THE SUM)where the constant radiant power is $\Phi = (a_0 A_0^{\mathrm{LG}} W_0)^2$, and $a_0 = \frac{\omega}{\sqrt{2c\mu_0}}$.

At first glance, the SOI appears to be absent, since ℓ and σ are not explicitly coupled in the second term of all three expressions. However, the trigonometric factor can be reinterpreted as

$$\sin\beta\cos\left[\frac{\pi}{2}(|\ell'|-|\ell|)+\alpha\right] = \left(\mathrm{Re}\left[e^{i(|\ell'|-|\ell|)}\right], \mathrm{Im}\left[e^{i(|\ell'|-|\ell|)}\right], 0\right)\cdot\begin{pmatrix}\sin\beta\cos\alpha \\ \sin\beta\sin\alpha \\ \cos\beta\end{pmatrix}, \tag{3.18}$$

which reveals its origin as a phase correction between the complex components of two VVB states located on the poles of Poincaré sphere.

More specifically, the complex phase of the radial factor $\tilde{F}_{\ell,p}(Q)$ takes the form $(-i)^{|\ell|}$. As a result, the interference is governed by the difference in $|\ell|$, which introduces a phase shift across whole momentum space, together with the VVB state specified by (α,β).

At the same time, the contribution from SOI (or, for BXs, the polarization-matching mechanism) is encoded through $\delta_{\Delta J,0}$. When $\Delta J \neq 0$, the resulting azimuthal phase mismatch—such as that illustrated in Figure **??**—leads to destructive averaging over momentum space, effectively given no net contribution to the total transition rate.

From this perspective, the factorial term in Eq. (3.17) can be interpreted as a geometric correction arising from the mismatch between the radial factors $\tilde{F}_{\ell,p}(Q)$ of the two LG beams. In particular, when $|\ell| = |\ell'|$, it reduces to simple constants, yielding 1 for BXs and $1+|\ell|$ for GXs. These values are consistent with the corresponding prefactors in the non-interference contribution (i.e., the term independent of $\delta_{\Delta J,0}$).

As for the beam waist W_0, which appears in the denominator of the prefactor for the GX total transition rate, it reflects that smaller W_0 enhances the longitudinal field component and thus increases the excitation efficiency of GXs.

On the other hand, the total transition rate for the total BXs under the condition $(\sigma' = -\sigma)$ can be similarly written as

$$\Gamma^{B\pm}_{\sigma,\ell,-\sigma,\ell'}(\alpha,\beta) \propto \sum_{S=B\pm} \left|\tilde{M}^{S}_{\sigma,\ell,-\sigma,\ell'}\right|^2 = \frac{\Phi}{4}\left(\frac{E_g D^B}{a_0 \Omega \hbar}\right)^2 . \tag{3.19}$$

Therefore, one can exploit both Eq. (3.17) and Eq. (3.19) to analyze the optical transition rates. In particular, we focus on the equator of the Poincaré sphere ($\beta = \pi/2$), where the VVB corresponds to a maximal superposition of two LG beams. By varying the azimuthal angle α along the equator, we examine the dependence of the total transition rate on ℓ (with $|\ell| = |\ell'|$) using the VVB states $|1, \ell, -1, -\ell; \alpha, \beta = \pi/2\rangle$, as shown in Figure 3.6.

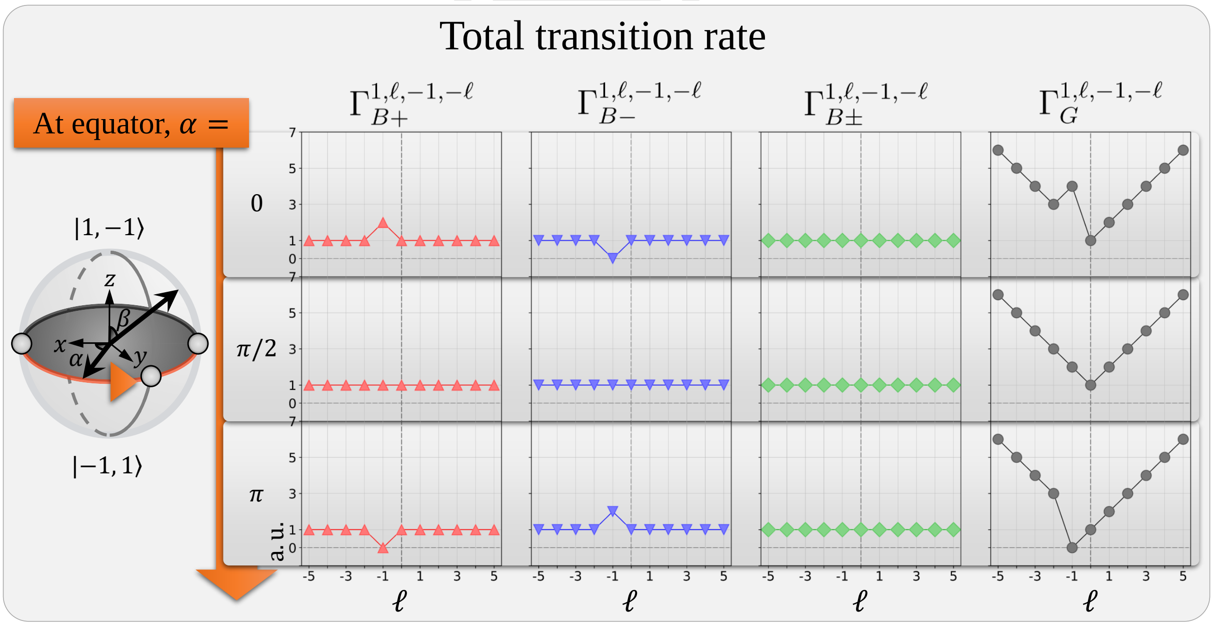

Figure 3.6 : Normalized total transition rates of BXs, total BXs, and GX excited by a VVB state $|1, \ell, -1, -\ell; \alpha, \beta = \pi/2\rangle$, for $\ell \in \mathbb{Z}$ with $\ell \in [-5, 5]$, and for azimuthal angles $\alpha = 0, \pi/2, \pi$.

In Figure 3.6, the total transition rate for each exciton exhibits a clear dependence on α when the condition $\Delta J = 0$ is satisfied at $\ell = -1$. As shown within the same column, varying α from 0 to $\pi/2$ to π leads to modulation of the transition rate. This behavior is consistent with the results in Figure 3.2 and Figure 3.5, where the transition rate exhibits fully constructive or destructive interference over momentum space at $\alpha = 0$ or $\alpha = \pi$.

In contrast, the total transition rate of total BXs becomes independent of α and ℓ. This is due to the cancellation between the contributions of the upper and lower BX bands, as reflected in Figure **??**.

Apart from this special cancellation, BXs generally exhibit independence from ℓ, unlike

GXs. This difference originates from the additional $\boldsymbol{Q}$-dependent prefactor in the GX case, specifically $\sin\theta_{\boldsymbol{Q}} \approx Q/q_0$ in Eq. (3.15). Consequently, Eq. (3.17) shows a dependence on ℓ even in the absence of interference effects.

Motivated by the special properties of the VVB state at the equator of the Poincaré sphere, $|1,-1,-1,1;\alpha,\beta=\pi/2\rangle$, one can achieve smooth and selective excitation over half of the equator ($\alpha=0$ to π), as shown in Figure 3.7.

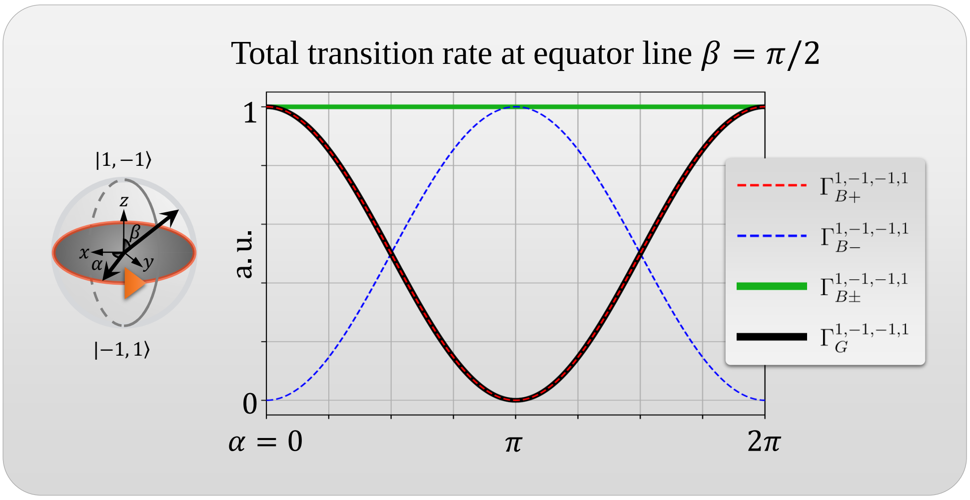

Figure 3.7：Normalized total transition rates of BXs, total BXs, and GXs excited by a VVB state $|1,-1,-1,1;\alpha,\beta=\pi/2\rangle$ for azimuthal angles $\alpha=0$ to 2π.

These observations originate from SOI associated with the longitudinal component of the LG beam, together with a polarization-matching mechanism. The former arises from the gauge transformation, specifically the second term in Eq. (2.23), as well as the intrinsic structure of the LG beam. The latter originates from the correspondence between the transition dipole structure of the upper and lower BX bands and the polarization textures of the VVB state. In both cases, Fermi's golden rule provides a theoretical framework for interpreting these results and establishes a basis for further studies involving different VVB configurations, excitonic species, and other comprehensive investigations.

Chapter 4 Wave Packets

4.1 Internal Structure of Excitons

4.1.1 Global Phase in Valley-Mixed Exciton

In this section, we establish the theoretical foundation for further studies of exciton wave packets. A wave packet of the particle refers to a superposition of momentum eigenstates, where the carrier wave (ripples) possesses a characteristic wavelength determined by the dominant momentum component, while the envelope describes the finite spatial extent of the particle. The velocity associated with the envelope, known as the group velocity, corresponds to the particle velocity in classical physics. From this perspective, the envelope of a wave packet provides a good intuition of the particle's position and motion [41, 43].

Using time-dependent perturbation theory [41], the time-dependent exciton state under weak photoexcitation can be expressed as a superposition of finite-momentum exciton (SFME) states [6],

$$|\Psi(t)\rangle = |GS\rangle + \sum_{S=B\pm} \sum_{\boldsymbol{Q}} \tilde{c}^{S,\boldsymbol{Q}(1)}(t) e^{-i\omega^{S,\boldsymbol{Q}}t} |\Psi^{S,\boldsymbol{Q}}\rangle, \tag{4.1}$$

which represents a linear combination of the ground state $|GS\rangle$ and exciton states $|\Psi^{S,\boldsymbol{Q}}\rangle$ with center-of-mass (COM) momentum $\boldsymbol{Q}$. Other notations follow Eq. (3.2). Within first-order perturbation theory, the amplitude coefficients are given by

$$\tilde{c}^{S,\boldsymbol{Q}(1)}(t) = \tilde{\gamma}^{S,\boldsymbol{Q}}(t) \left(\frac{\tilde{M}^{S,\boldsymbol{Q}}}{\hbar\omega^{S,\boldsymbol{Q}}} \right), \tag{4.2}$$

where time–dependent coefficient $\tilde{\gamma}^{S,\boldsymbol{Q}}(t) \approx \tilde{\gamma}^{\boldsymbol{0}}(t)$, and the natural frequency is approximated as $\omega^{S,\boldsymbol{Q}} = E^{S,\boldsymbol{Q}}/\hbar \approx E^{S,\boldsymbol{0}}/\hbar = \omega^{\boldsymbol{0}}$. The amplitude coefficients can thus be separated into time–dependent and time–independent parts, where the latter contains the optical matrix element introduced in Eq. (3.8). For simplicity, we consider excitation by a single LG beam with radial mode index $p = 0$, such that the optical matrix element takes the form

$$\tilde{M}^{S,\boldsymbol{Q}}_{\hat{\varepsilon},\ell} \approx \frac{E_g}{2i\hbar} \left(\boldsymbol{\mathcal{A}}_{\hat{\varepsilon},\ell}(\boldsymbol{Q}) \cdot \boldsymbol{D}^{S,\boldsymbol{Q}*} \right). \tag{4.3}$$

Here, derivations analogous to those in the previous chapter—including the rotating-wave approximation and electric dipole approximation—are omitted for brevity. By adopting the polarization unit vector $\hat{\varepsilon}$ instead of photon helicity σ, we relax the restriction to circular polarization. The vector potential is denoted by $\boldsymbol{\mathcal{A}}_{\hat{\varepsilon},\ell}(\boldsymbol{q}_{\parallel})$, which corresponds to the complex amplitude introduced in Eq. (2.27) but with polarization unit vector not limited to circularly polarized.

In the work of Guan-Hao Peng *et al.* [6], the envelope function is defined as

$$F^S_{\hat{\varepsilon},\ell}(\boldsymbol{R}_c, t) = \frac{1}{\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{c}^{S,\boldsymbol{Q}(1)}(t) e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}. \tag{4.4}$$

To explicitly isolate the static contribution, we use the relation $F^S_{\hat{\varepsilon},\ell}(\boldsymbol{R}_c, t) \approx \tilde{\gamma}_{\mathbf{0}}(t)\, f^S_{\hat{\varepsilon},\ell}(\boldsymbol{R}_c)$, which defines the static envelope function as

$$f^S_{\hat{\varepsilon},\ell}(\boldsymbol{R}_c) = \frac{1}{\hbar\omega^{\mathbf{0}}\sqrt{\Omega}} \sum_{\boldsymbol{Q}} \tilde{M}^{S,\boldsymbol{Q}}_{\hat{\varepsilon},\ell} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}. \tag{4.5}$$

However, the valley-mixed exciton state obtained from the pseudo-spin model [6],

$$|\Psi^{\pm,\boldsymbol{Q}}\rangle = e^{i\delta(\boldsymbol{Q})} \left(e^{-i\phi_{\boldsymbol{Q}}} |\Psi^{K,\boldsymbol{Q}}\rangle \pm e^{i\phi_{\boldsymbol{Q}}} |\Psi^{K',\boldsymbol{Q}}\rangle \right) /\sqrt{2}, \tag{4.6}$$

naturally carries a $\boldsymbol{Q}$-dependent global phase $\delta(\boldsymbol{Q})$ arising from its internal structure. Consequently, the optical matrix element acquires the same phase factor,

$$\tilde{M}^{\pm,\boldsymbol{Q}}_{\hat{\varepsilon},\ell} = e^{i\delta(\boldsymbol{Q})} \left(e^{-i\phi_{\boldsymbol{Q}}} \tilde{M}^{K,\boldsymbol{Q}}_{\hat{\varepsilon},\ell} \pm e^{i\phi_{\boldsymbol{Q}}} \tilde{M}^{K',\boldsymbol{Q}}_{\hat{\varepsilon},\ell} \right) /\sqrt{2}. \tag{4.7}$$

As a result, this $\boldsymbol{Q}$-dependent phase can modify the resulting static envelope function across momentum space, provided that the internal phase factors $(e^{\pm i\phi_{\boldsymbol{Q}}})$ partially compensate or reshape the global phase. This observation motivates a more careful examination of the internal structure of the exciton wave function and the explicit extraction of common factors.

4.2 Wave functions of finite-momentum excitons

To address this issue without resorting to a numerically expensive construction of the exciton wave function, we instead extract its $\boldsymbol{Q}$–dependent phase structure analytically. We begin by considering the internal structure of the exciton wave function [44, 45],

$$\Psi^{S,\boldsymbol{Q}}(\boldsymbol{r}_e, \boldsymbol{r}_h) = \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k}) \psi_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{r}_e) \psi^*_{v,\boldsymbol{k}}(\boldsymbol{r}_h), \tag{4.8}$$

where $\psi_{c,\boldsymbol{k}}(\boldsymbol{r}_e)$ and $\psi_{v,\boldsymbol{k}}(\boldsymbol{r}_h)$ denote the single-particle Bloch wave functions for an electron in the conduction band and a hole in the valence band, respectively. Here, $\boldsymbol{r}_e$ and $\boldsymbol{r}_h$ are the corresponding electron and hole position vectors. The coefficient $\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k})$ represents the amplitude of the electron–hole configuration for exciton band S.

For the purpose of extracting the COM momentum $\boldsymbol{Q}$-dependent phase term, it is convenient to rewrite the exciton wave function in terms of a weighted-average position vector $\boldsymbol{R}_c$[1]

[1] Although this weighted-average position vector is not strictly identical to the conventional COM coordinate in classical mechanics, we retain the subscript "c" to avoid confusion with the Bravais lattice vector $\boldsymbol{R}$.

and the relative coordinate $\boldsymbol{r}$:

$$\begin{aligned} \boldsymbol{R}_c = a\boldsymbol{r}_e + b\boldsymbol{r}_h, \qquad \boldsymbol{r} = \boldsymbol{r}_e - \boldsymbol{r}_h, \qquad a + b = 1, \\ \boldsymbol{r}_e = \boldsymbol{R}_c + b\boldsymbol{r}, \qquad \boldsymbol{r}_h = \boldsymbol{R}_c - a\boldsymbol{r}. \end{aligned} \tag{4.9}$$

Using these relations, the exciton wave function can be rewritten as

$$\Psi^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) = \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k}) \psi_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r}) \psi^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c - a\boldsymbol{r}). \tag{4.10}$$

In addition, Bloch's theorem states that, in a material with a periodic lattice structure—such as a two-dimensional TMD characterized by Bravais lattice vectors $\boldsymbol{R}$—the single-particle wave function satisfies [41, 45]

$$\psi_{n,\boldsymbol{k}}(\boldsymbol{r}') = u_{n,\boldsymbol{k}}(\boldsymbol{r}') e^{i\boldsymbol{k}\cdot\boldsymbol{r}'}, \tag{4.11}$$

where the wave function is expressed as the product of a cell-periodic function, $u_{n,\boldsymbol{k}}(\boldsymbol{r}') = u_{n,\boldsymbol{k}}(\boldsymbol{r}' + \boldsymbol{R})$, and a plane-wave component $e^{i\boldsymbol{k}\cdot\boldsymbol{r}'}$ [2]. Using this property, the exciton wave function can be rewritten as

$$\begin{aligned} &\Psi^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) \\ &\Rightarrow \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k} - b\boldsymbol{Q}) \psi_{c,\boldsymbol{k}+a\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r}) \psi^*_{v,\boldsymbol{k}-b\boldsymbol{Q}}(\boldsymbol{R}_c - a\boldsymbol{r}) \\ &= \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k} - b\boldsymbol{Q}) \left[u_{c,\boldsymbol{k}+a\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r}) e^{i(\boldsymbol{k}+a\boldsymbol{Q})\cdot(\boldsymbol{R}_c+b\boldsymbol{r})} u^*_{v,\boldsymbol{k}-b\boldsymbol{Q}}(\boldsymbol{R}_c - a\boldsymbol{r}) e^{-i(\boldsymbol{k}-b\boldsymbol{Q})\cdot(\boldsymbol{R}_c-a\boldsymbol{r})} \right] \\ &= \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k} - b\boldsymbol{Q}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} \left[u_{c,\boldsymbol{k}+a\boldsymbol{Q}}(\boldsymbol{R}_c + b\boldsymbol{r}) u^*_{v,\boldsymbol{k}-b\boldsymbol{Q}}(\boldsymbol{R}_c - a\boldsymbol{r}) \right]. \\ &= e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) \end{aligned} \tag{4.12}$$

The first arrow is obtained by shifting crystal momentum by $-b\boldsymbol{Q}$, which allows the $\boldsymbol{Q}$-dependent phase originating from the plane-wave components to be written in the compact form $e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}$. Finally, we extract $U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r})$. Once we eliminate the $\boldsymbol{Q}$-dependence of $U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r})$, it should be easier to construct the internal structure of the exciton wave function

Finally, we adopt the convention

$$a = 1, \qquad b = 0,$$

then

$$U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c, \boldsymbol{r}) = \sum_{vc\boldsymbol{k}} \Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \left[u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c) u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c - \boldsymbol{r}) \right]. \tag{4.13}$$

Under this convention, the whole cell-periodic function becomes independent of $\boldsymbol{Q}$.

Further simplification can be achieved by performing a Taylor expansion around $\boldsymbol{Q} = \boldsymbol{0}$

[2] The vector $\boldsymbol{r}'$ denotes a general position vector and may represent the weighted-average, relative, electron, or hole position vector.

for the implicitly $\boldsymbol{Q}$-dependent quantities, namely $\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k})$ and $u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c)$. Explicitly,

$$
\begin{aligned}
\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k}) &= \sum_{m=0}^{\infty} \frac{1}{m!} \left(\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\right)^m \Lambda_{S,\mathbf{0}}(vc\boldsymbol{k}) &&= \Lambda_{S,\mathbf{0}}(vc\boldsymbol{k}) + \boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\Lambda_{S,\mathbf{0}}(vc\boldsymbol{k}) + \cdots, \\
u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c) &= \sum_{m=0}^{\infty} \frac{1}{m!} \left(\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\right)^m u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) &&= u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) + \boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}} u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) + \cdots.
\end{aligned}
\tag{4.14}
$$

Their product can therefore be expanded as

$$
\begin{aligned}
\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k}) u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c) = &\underbrace{\Lambda^{S,\mathbf{0}}(vc\boldsymbol{k}) u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)}_{\text{zeroth order}} \\
&+ \underbrace{\Lambda^{S,\mathbf{0}}(vc\boldsymbol{k}) \left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}} u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)\right] + \left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\Lambda^{S,\mathbf{0}}(vc\boldsymbol{k})\right] u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)}_{\text{first order}} \\
&+ \underbrace{\left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}}\Lambda^{S,\mathbf{0}}(vc\boldsymbol{k})\right] \left[\boldsymbol{Q}\cdot\boldsymbol{\nabla}_{\boldsymbol{k}} u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)\right]}_{\text{second order}} \\
&+ \text{higher-order terms.}
\end{aligned}
\tag{4.15}
$$

4.3 Zeroth-order approximation

Substituting this expansion into Eq. (4.13), the exciton wave function can be decomposed into contributions from different orders of $\boldsymbol{Q}$,

$$
\begin{aligned}
U^{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) &= \sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \left\{\Lambda^{S,\boldsymbol{Q}}(vc\boldsymbol{k}) u_{c,\boldsymbol{k}+\boldsymbol{Q}}(\boldsymbol{R}_c)\right\} u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c-\boldsymbol{r}) \\
&= \sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \left[\Psi^{S,\boldsymbol{Q}(0)}(\boldsymbol{R}_c,\boldsymbol{r}) + \Psi^{S,\boldsymbol{Q}(1)}(\boldsymbol{R}_c,\boldsymbol{r}) + \Psi^{S,\boldsymbol{Q}(2)}(\boldsymbol{R}_c,\boldsymbol{r}) + \cdots\right] \\
&\approx \sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \Lambda^{S,\mathbf{0}}(vc\boldsymbol{k}) u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c-\boldsymbol{r}) \quad \text{———} \quad \text{zeroth–order},
\end{aligned}
\tag{4.16}
$$

where the final approximation refers to the zeroth-order approximation [3] in the small-$\boldsymbol{Q}$ limit. Consequently, the $\boldsymbol{Q}$-dependent phase factor can be extracted as [46]

$$
\Psi^{S,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \approx e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} U^{S,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})
\tag{4.17}
$$

In addition, the valley exciton wave functions with zero COM momentum satisfy

$$
U^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}) = U^{K',\mathbf{0}*}(\boldsymbol{R}_c,\boldsymbol{r})
\tag{4.18}
$$

[3] A similar expansion scheme is commonly employed in evaluating the electron–hole exchange interaction (EHEI) within the Bethe–Salpeter equation, where the first-order contribution is typically retained. Although this may appear inconsistent with the zeroth-order approximation adopted here, the zeroth-order contribution to the EHEI vanishes to leading order, thereby necessitating the inclusion of higher-order terms.

as derived in appendix A.3. Therefore, each wave function can be expressed in polar form as

$$\begin{aligned} U^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}) &= \left|U^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right| \exp\left[i\,\mathrm{Arg}\big(U^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\big)\right], \\ U^{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}) &= \left|U^{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right| \exp\left[i\,\mathrm{Arg}\big(U^{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\big)\right]. \end{aligned} \tag{4.19}$$

From the complex-conjugation relation above, the two wave functions have identical amplitudes while their phases are opposite in sign, i.e.,

$$\left|\Psi^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right| = \left|\Psi^{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right|, \qquad \mathrm{Arg}\left[\Psi^{K,\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right] = -\mathrm{Arg}\left[\Psi^{K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right]. \tag{4.20}$$

4.4 Exciton Wave Packet for Excitons

4.4.1 Valley Excitons

First, for simplicity we disregard the electron-hole exchange interaction and exciton states theat possess well-defoned valley degree of freedom, K or K'. In the zeroth-order approximation, Eq. (4.17) suggests

$$\Psi^{K/K',\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \approx e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} U^{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}). \tag{4.21}$$

Accordingly, considering the internal structure of the exciton states, the envelope function is constructed as a weighted superposition of exciton momentum eigenstates in real space, with amplitude coefficients derived from first–order time–dependent perturbation theory as

$$\begin{aligned} \Phi^{K/K'}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,\boldsymbol{r},t) &= \sum_{\boldsymbol{Q}} \tilde{c}^{K/K',\boldsymbol{Q}(1)}(t) e^{-i\omega^{K/K',\boldsymbol{Q}} t} \Psi^{K/K',\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \\ &\approx U^{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}) \sum_{\boldsymbol{Q}} \tilde{c}^{K/K',\boldsymbol{Q}(1)}(t) e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}, \end{aligned} \tag{4.22}$$

where the internal structure of the zero COM exciton wave function is given by Eq. (4.13) as

$$U^{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r}) = \sum_{vc\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \Lambda^{S,\mathbf{0}}(vc\boldsymbol{k}) u_{c,\boldsymbol{k}}(\boldsymbol{R}_c) u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c-\boldsymbol{r}). \tag{4.23}$$

At this stage, both $U^{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})$ and the time–dependent coefficient $\tilde{c}^{K/K',\boldsymbol{Q}(1)}(t)$ arising from the definition of the envelope function prevents further simplification. In addition, $U^{K/K',\mathbf{0}}(\boldsymbol{R}_c,\boldsymbol{r})$ contain three more components : the phase factor $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$, the coefficient $\Lambda^{S,\mathbf{0}}(vc\boldsymbol{k})$, and the cell-periodic functions $u_{c,\boldsymbol{k}}(\boldsymbol{R}_c)$ and $u^*_{v,\boldsymbol{k}}(\boldsymbol{R}_c-\boldsymbol{r})$. In the following, we address this issue by introducing physically motivated conditions under which its contribution can be regarded as negligible.

Time–dependent coefficient

Same as Eq. (4.2), the time–dependent coefficient can be written and approximated as

$$\tilde{c}^{K/K',\boldsymbol{Q}(1)}(t) = \frac{\tilde{\gamma}^{K/K',\boldsymbol{Q}}(t)}{\hbar\omega^{K/K',\boldsymbol{Q}}}\tilde{M}^{S,\boldsymbol{Q}} \approx \frac{\tilde{\gamma}^{\boldsymbol{0}}(t)}{\hbar\omega^{\boldsymbol{0}}}\tilde{M}^{K/K',\boldsymbol{Q}}. \tag{4.24}$$

The remaining time–dependent part is disregard for now. Thus, the optical matrix elements will be the main component in changing the shape of the wave packet.

The phase factor $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$

If we focusing on the case where the electron and hole spatially overlap, i.e., $\boldsymbol{r} = \boldsymbol{0}$, the phase factor $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$ can be safely neglected . Beyond simplifying the analysis, this constraint physically corresponds to the configuration required for an exciton to undergo radiative recombination.

The coefficient $\Lambda^{S,\boldsymbol{0}}(vc\boldsymbol{k})$

Second, the variation of the coefficient $\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k})$ with respect to $\boldsymbol{k}$ can be estimated using the two-dimensional hydrogen model [47]. For the $1s$ exciton state, it is given by

$$\Lambda^{1s,\boldsymbol{0}}(vc\boldsymbol{k}) = \frac{\sqrt{2\pi}a_B^*}{\left[1 + \left(\frac{a_B^*\boldsymbol{k}}{2}\right)^2\right]^{3/2}}, \tag{4.25}$$

where the effective Bohr radius is $a_B^* \approx 1\,\text{nm}$. As shown in Figure 4.1, the coefficient $\Lambda_{S,\boldsymbol{0}}(vc\boldsymbol{k})$ remains nearly constant within the light cone region $\boldsymbol{Q}_c$, and is therefore expected to have a negligible influence on the summation.

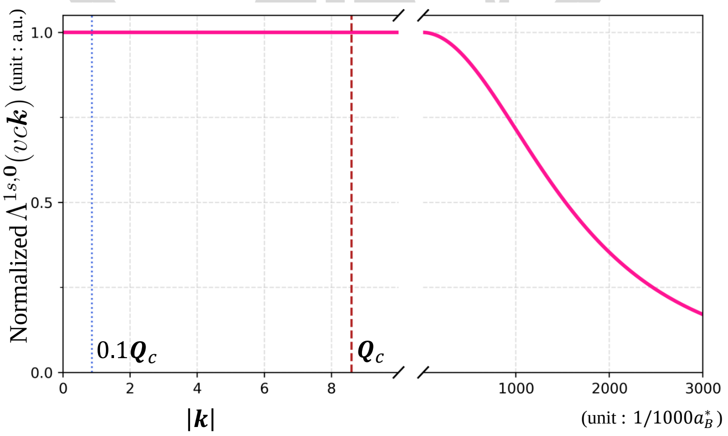

Figure 4.1 : Normalized coefficient $\Lambda^{S,\boldsymbol{0}}(vc\boldsymbol{k})$ along $\boldsymbol{k}$ (magenta), where the vertical dashed lines indecate the 0.1 and 1 times light cone range $\boldsymbol{Q}_c$ (blue and red). The light cone range $\boldsymbol{Q}_c = 8.6\mu\text{m}^{-1}$ is considered for WSe_2. The maximun value is corresponding to $\sqrt{2\pi}a_B^*$

The cell-periodic functions

Finally, the cell-periodic functions satisfy $u_{n,\boldsymbol{k}}(\boldsymbol{R}_c) = u_{n,\boldsymbol{k}}(\boldsymbol{R}_c + \boldsymbol{R})$. Since the lattice constant (on the order of Å) is much smaller than the optical beam waist ($W_0 \sim 1.5\,\mu\text{m}$), the variation of the cell-periodic functions is confined within a unit cell and does not significantly affect the overall spatial profile. From another perspective, neglecting the cell-periodic functions corresponds to describing the envelope function at discrete lattice sites, which does not qualitatively alter the result.

With the condition obtained above, and adopting Born's statistical interpretation [41], the probability density can be estimated from the absolute square of the static envelope function. In particular, for $\boldsymbol{r} = \boldsymbol{0}$,

$$\left|\Phi_{\hat{\varepsilon},\ell}^{K/K'}(\boldsymbol{R}_c, \boldsymbol{r} = \boldsymbol{0})\right|^2 \propto \left|\sum_{\boldsymbol{Q}} \tilde{M}_{\hat{\varepsilon},\ell}^{K/K',\boldsymbol{Q}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\right|^2 . \tag{4.26}$$

Simulations

We begin with a brief summary to following simulations. For valley excitons, only the transverse component of light contributes to the optical matrix element, yielding

$$\tilde{M}_{\sigma,\ell}^{S,\boldsymbol{Q}} \approx \frac{E_g}{2i\hbar}\left(\boldsymbol{\mathcal{A}}_{\hat{\varepsilon},\ell}^{\parallel}(\boldsymbol{Q}) \cdot \boldsymbol{D}_{S,\boldsymbol{Q}}^{X*}\right). \tag{4.3}$$

The transverse component of the vector potential in the Coulomb gauge is given in Eq. (2.27). Accordingly, it can be written as

$$\boldsymbol{\mathcal{A}}_{\hat{\varepsilon},\ell}^{\parallel}(\boldsymbol{Q}) = \mathcal{A}_{\ell}^{\parallel}(\boldsymbol{Q})\hat{\varepsilon} \approx \tilde{F}_{\ell,p=0}(\boldsymbol{Q})e^{i\ell\phi_{\boldsymbol{Q}}}\hat{\varepsilon}, \tag{4.27}$$

where the approximation follows from the paraxial condition adopted in Eq. (2.30).

Thus, the $\boldsymbol{Q}$-dependent phase is modified through the dot product between light and the transition dipole moment and eventually captures the spatial profile of the exciton probability distribution. The following simulations illustrate the squared magnitude distributions of the transverse component of the LG beam, $\left|\mathcal{A}_{\ell}^{\parallel}(\boldsymbol{Q})\right|^2$, the optical matrix element, $\left|\tilde{M}_{\sigma,\ell}^{S,\boldsymbol{Q}}\right|^2$, and the resulting static envelope function, $\left|\Phi_{\hat{\varepsilon},\ell}^{K/K'}(\boldsymbol{R}_c, \boldsymbol{r} = \boldsymbol{0})\right|^2$ In this way, we can track how the spatial profile evolves across these stages and relate each feature to its underlying physical origin.

For topological charge $\ell = 0$ and ± 1, the Gaussian–like and doughnut–shaped structures (see Figure 4.2 and Figure 4.4) are preserved in both the optical matrix element and the resulting probability distribution (see Figure 4.3 and Figure 4.5). This behavior directly reflects the spatial structure of the LG beam in angular spectrum space, which exhibits corresponding profiles in real space as well (see Figure 2.4).

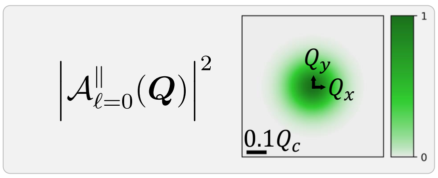

Figure 4.2：Normalized squared magnitude distribution of the transverse component of the LG beam ($\ell = 0$). The scale bar indicates 0.1 times the light cone range for WSe_2 monolayers, $\boldsymbol{Q}_c = 8.6\ \mu\text{m}^{-1}$.

In addition, the effect of strong spin–orbit coupling manifests as valley–selective optical transitions, which are also evident here: K and K' valley excitons are selectively excited by circularly polarized light with $\hat{\varepsilon} = \hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}}$ and $\hat{\varepsilon} = \hat{\boldsymbol{x}} - i\hat{\boldsymbol{y}}$, respectively.

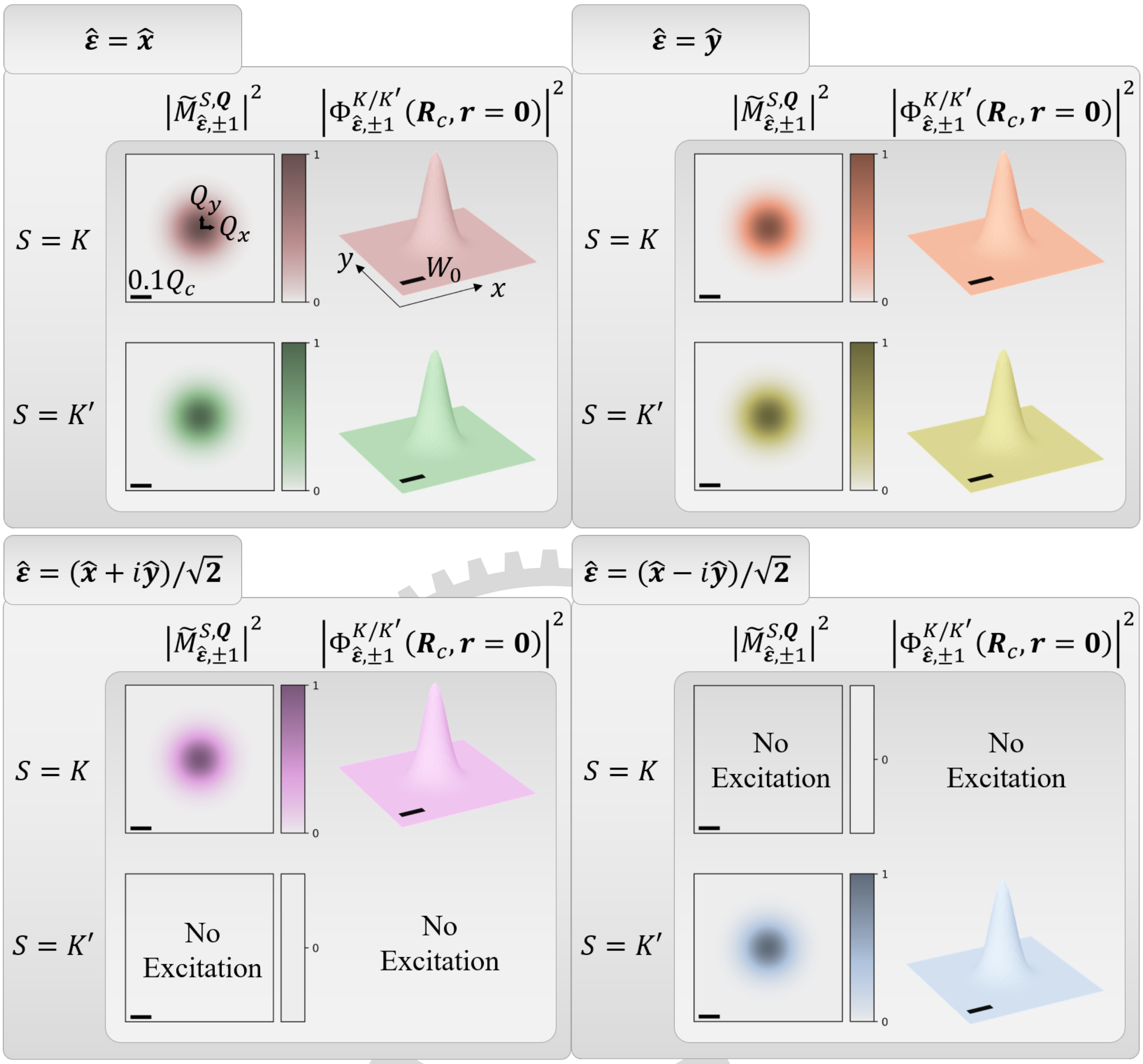

Figure 4.3：Normalized squared magnitudes of the optical matrix element and the valley exciton wave packet excited by a LG beam ($\ell = 0$) with different polarization unit vectors $\hat{\varepsilon}$. The scale bar indicates $0.1\,\boldsymbol{Q}_c$, and the beam waist is $W_0 = 1.5\,\mu\text{m}^{-1}$.

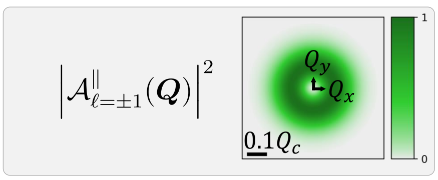

Figure 4.4：Normalized squared magnitude distribution of the transverse component of the LG beam ($\ell = 1$). The scale bar indicates 0.1 times the light cone range for WSe_2 monolayers, $\boldsymbol{Q}_c = 8.6\,\mu\text{m}^{-1}$.

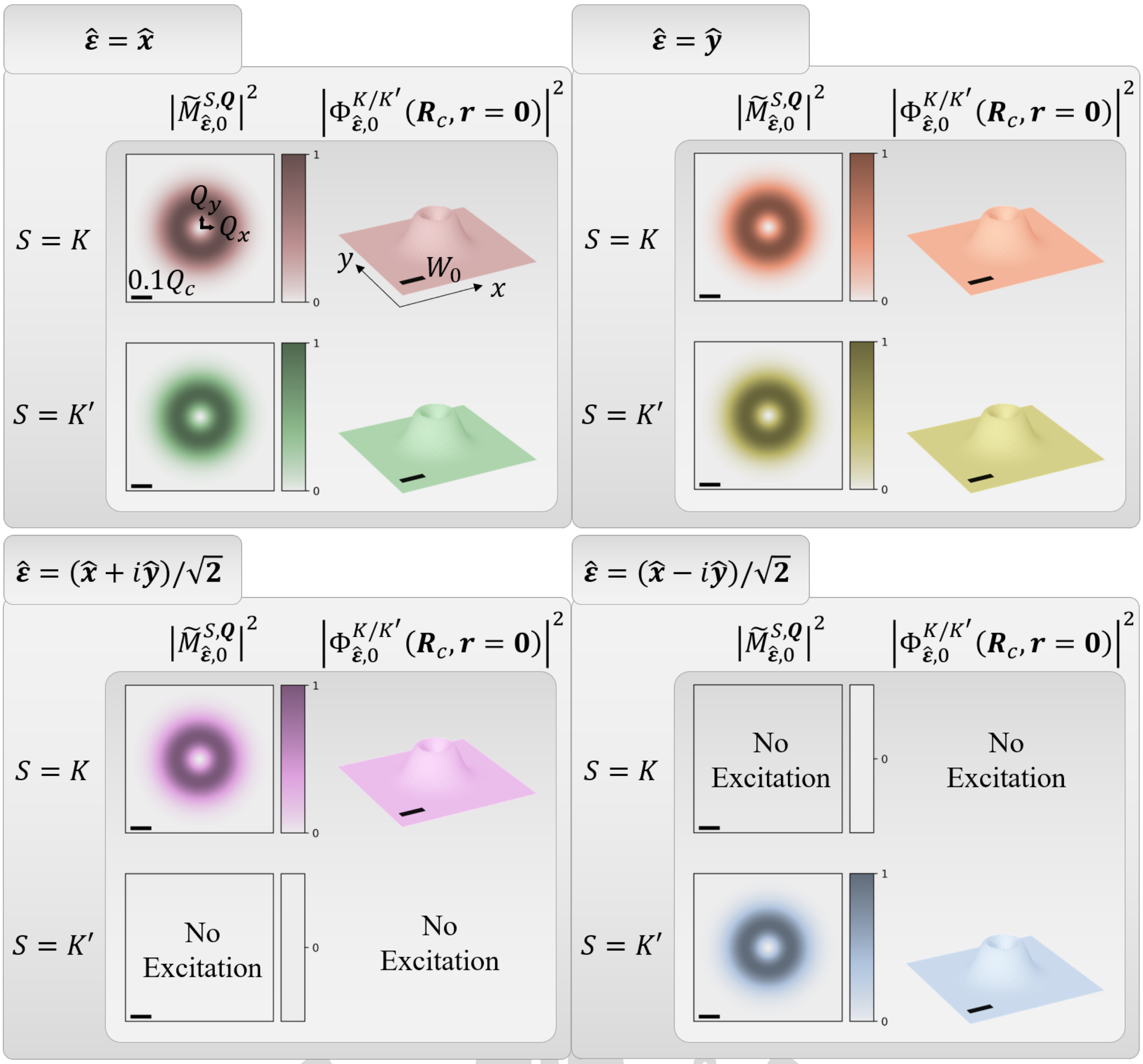

Figure 4.5 : Normalized squared magnitudes of the optical matrix element and the valley exciton wave packet excited by a LG beam ($\ell = 1$) with different polarization unit vectors $\hat{\varepsilon}$. The scale bar indicates $0.1\,\boldsymbol{Q}_c$, and the beam waist is $W_0 = 1.5\,\mu\text{m}^{-1}$.

It is also interesting to look at the case $\Phi^{K/K'}_{\hat{\varepsilon},\ell}(\boldsymbol{R}_c, \boldsymbol{r} \neq \boldsymbol{0})$. For example, the separation of electron and hole equals to the Bohr radius ($|\boldsymbol{r}| = a_{\mathrm{B}}$). However, two of the simplifications that have been applied on the valley exciton wave functions with zero COM momentum $U^{K/K',\boldsymbol{0}}(\boldsymbol{R}_c, \boldsymbol{r})$ would fail. First, the phase factor $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$ will become non–zero. Second, the $\boldsymbol{r}$–dependence in the cell-periodic functions for holes. Thus we will need to trace back to Eq. (4.23) instead of Eq. (4.26)

4.4.2 Valley-mixed Excitons

For valley–mixed excitons, the corresponding wave functions can be constructed within the pseudo–spin model [6] from the valley exciton wave functions. Notably, Eq. (4.17) remains applicable in this case. One obtains

$$\begin{aligned}\Psi^{\pm,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) &\approx \frac{1}{\sqrt{2}}\left[e^{-i\phi_{\boldsymbol{Q}}}\Psi^{K,\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r}) \pm e^{i\phi_{\boldsymbol{Q}}}\Psi^{K',\boldsymbol{Q}}(\boldsymbol{R}_c,\boldsymbol{r})\right] \\ &\approx e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}U^{\pm,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}).\end{aligned} \tag{4.28}$$

Following Eq. (4.22) and combining it with Eq. (4.28), the envelope function can be written as

$$\Phi^{\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,\boldsymbol{r},t) = \sum_{\boldsymbol{Q}} \tilde{c}^{\pm,\boldsymbol{Q}(1)}(t)e^{-i\omega_{\pm,\boldsymbol{Q}}t}U^{\pm,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}). \tag{4.29}$$

Similarly, approximate the time–dependent coefficient with Eq. (4.24) the static part of envelope function is written as

$$\Phi^{\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,\boldsymbol{r}) = \left(\hbar\omega^{\boldsymbol{0}}\right)^{-1}\sum_{\boldsymbol{Q}} \tilde{M}^{\pm,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}U^{\pm,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}). \tag{4.30}$$

Here, $U^{\pm,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})$ is retained inside the summation because the $\phi_{\boldsymbol{Q}}$ dependence remains embedded in its internal structure. Using both Eq. (4.6) and Eq. (4.7), the $\boldsymbol{Q}$–dependent terms can be written more explicitly by rewriting the static envelope function in terms of the valley indices K and K':

$$\begin{aligned}&\Phi^{\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,\boldsymbol{r}) \\ &= \left(2\hbar\omega^{\boldsymbol{0}}\right)^{-1}\sum_{\boldsymbol{Q}}\left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{K,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}\right)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}\left[e^{-i\phi_{\boldsymbol{Q}}}U^{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r}) \pm e^{i\phi_{\boldsymbol{Q}}}U^{K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right] \\ &= \left(2\hbar\omega^{\boldsymbol{0}}\right)^{-1}\left[U^{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\sum_{\boldsymbol{Q}}\left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{K,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}\right)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}e^{-i\phi_{\boldsymbol{Q}}}\right. \\ &\qquad\left.\pm\, U^{K',\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\sum_{\boldsymbol{Q}}\left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{K,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}\right)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}e^{i\phi_{\boldsymbol{Q}}}\right].\end{aligned} \tag{4.31}$$

Furthermore, using the complex–conjugation relation in Eq. (4.20), the amplitudes of the wave functions can be extracted, yielding

$$\begin{aligned}\Phi^{\pm}_{\hat{\boldsymbol{\varepsilon}},\ell}(\boldsymbol{R}_c,\boldsymbol{r}) = \left(2\hbar\omega^{\boldsymbol{0}}\right)^{-1}&\left|U^{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right| \\ &\left[e^{i\mathrm{Arg}\left[U^{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right]}\sum_{\boldsymbol{Q}}\left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{K,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}\right)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}e^{-i\phi_{\boldsymbol{Q}}}\right. \\ &\left.\pm\, e^{-i\mathrm{Arg}\left[U^{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right]}\sum_{\boldsymbol{Q}}\left(e^{-i\phi_{\boldsymbol{Q}}}\tilde{M}^{K,\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell} \pm e^{i\phi_{\boldsymbol{Q}}}\tilde{M}^{K',\boldsymbol{Q}}_{\hat{\boldsymbol{\varepsilon}},\ell}\right)e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c}e^{i\phi_{\boldsymbol{Q}}}\right].\end{aligned} \tag{4.32}$$

In this form, the phase factors $\left(e^{\pm i\mathrm{Arg}\left[U^{K,\boldsymbol{0}}(\boldsymbol{R}_c,\boldsymbol{r})\right]}\right)$ can no longer be neglected. This is because the summations explicitly depend on the $\boldsymbol{Q}$-dependent terms $e^{\pm i\phi_{\boldsymbol{Q}}}$, which originate from the construction of the valley–mixed exciton states and give rise to nontrivial interference effects in momentum space. As a result, a complete construction of the exciton wave function appears, at first sight, unavoidable.

As a correspondence to the each terms appearing in Eq. (4.32), an artificial phase term could be added through

$$\left|\Phi_{\hat{\varepsilon},\ell}^{K/K'}(\boldsymbol{R}_c, \boldsymbol{r}=\boldsymbol{0})\right|^2 \propto \left|\sum_{\boldsymbol{Q}} \tilde{M}_{\hat{\varepsilon},\ell}^{K/K',\boldsymbol{Q}} e^{i\boldsymbol{Q}\cdot\boldsymbol{R}_c} e^{-i\phi_{\boldsymbol{Q}}}\right|^2 \tag{4.33}$$

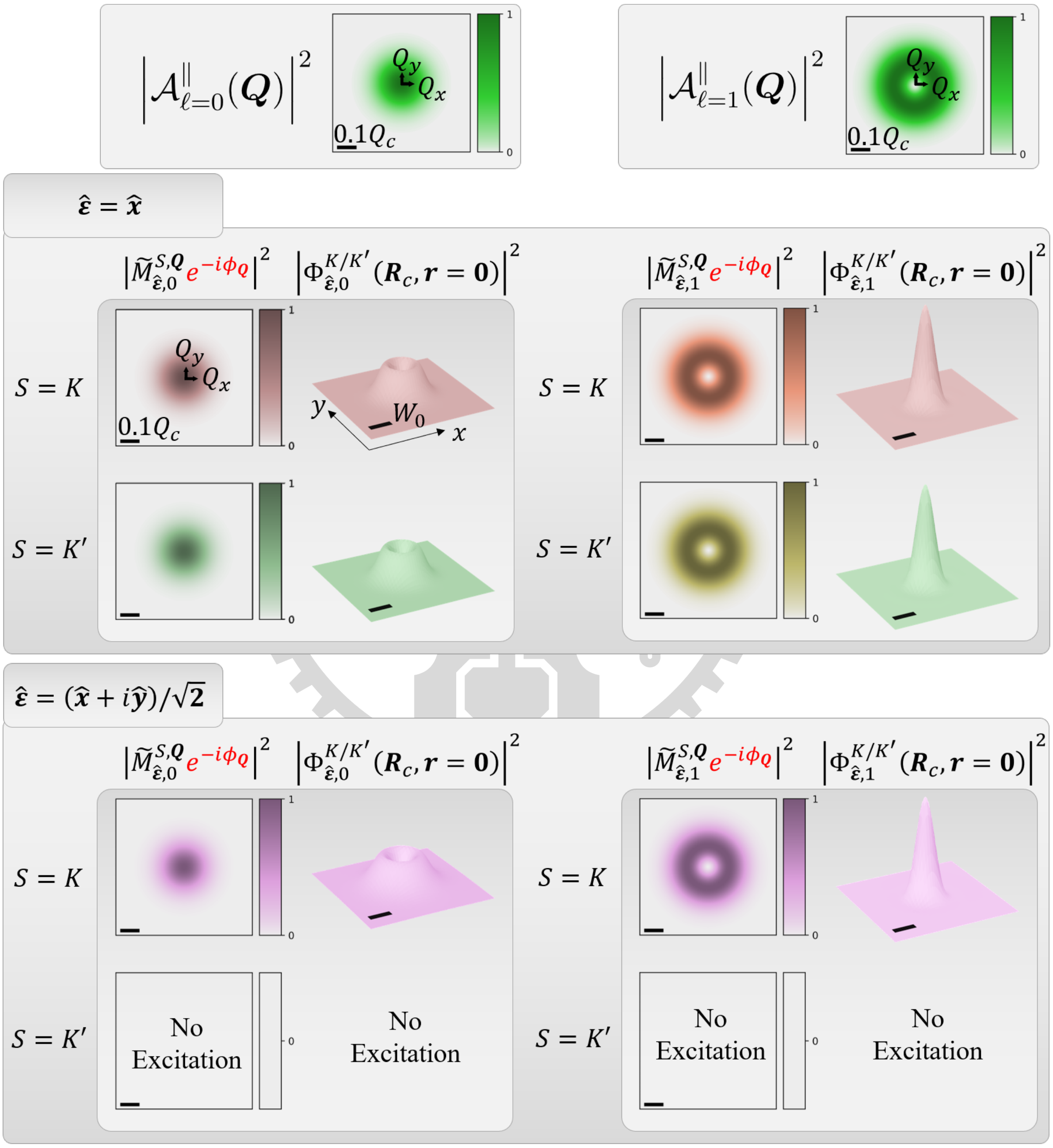

Figure 4.6 : Normalized squared magnitude distributions of the transverse component of the LG beam ($\ell = 0$ and 1), together with the normalized squared magnitudes of the optical matrix element and the resulting valley exciton wave packet for different polarization unit vectors $\hat{\varepsilon}$. The scale bar indicates 0.1 times the light cone range for WSe_2 monolayers, $\boldsymbol{Q}_c = 8.6\ \mu m^{-1}$, and the beam waist is $W_0 = 1.5\ \mu m^{-1}$.

The estimation of whether a peak appears at the center can be obtained as

$$
\begin{aligned}
\left|\Phi_{\hat{\boldsymbol{\varepsilon}},\ell}^{K/K'}(\boldsymbol{R}_c=\boldsymbol{0},\boldsymbol{r}=\boldsymbol{0})\right|^2 &\propto \left|\sum_{\boldsymbol{Q}} \tilde{M}_{\hat{\boldsymbol{\varepsilon}},\ell}^{K/K',\boldsymbol{Q}} e^{-i\phi_{\boldsymbol{Q}}}\right|^2 \\
&\propto \int_0^\infty \tilde{F}_{\ell,p=0}(Q)\left(\int_0^{2\pi} e^{i\ell\phi_{\boldsymbol{Q}}} e^{-i\phi_{\boldsymbol{Q}}}\, d\phi_{\boldsymbol{Q}}\right) dQ \\
&\propto \delta_{\ell,1},
\end{aligned} \tag{4.34}
$$

which serves as a reference for the subsequent simulations. For simplicity, we consider LG beams with polarization unit vectors $\hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{\varepsilon}} = (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}})/\sqrt{2}$. Similar behavior is observed for other polarization choices. In this case, the Gaussian–like and doughnut–shaped structures are no longer preserved in the resulting static envelope functions (see Figure 4.6). Instead, the profiles are effectively interchanged: a Gaussian–like beam excites a doughnut–shaped distribution, and vice versa. This qualitative change highlights the crucial role of the phase factors in Eq. (4.6), as the final summation is highly sensitive to these phase contributions.

Nevertheless, the valley exciton envelope functions obtained in Eq. (4.26) still capture essential features and exhibit a clear correspondence to the four sum terms appearing in Eq. (4.32). Motivated by this observation, in the remainder of this work we focus on simulating the valley exciton envelope functions and analyzing their characteristic features.

Chapter 5 Summary and outlook

In this study, we present a theoretical investigation of excitonic transitions in two–dimensional transition–metal dichalcogenides under photoexcitation by scalar and vector vortex beams.

In Chapter 2, we show that the well-defined orbital angular momentum (OAM) of $\ell\hbar$ carried by Laguerre–Gaussian (LG) beams is preserved under the paraxial approximation. Upon transformation to the Coulomb gauge, coupling with spin angular momentum (SAM) $\sigma\hbar$ naturally emerges. The adoption of the Coulomb gauge is justified within the framework of light–matter interaction, and the resulting spin–orbit interaction (SOI) term, $\sigma + \ell$, appears in the phase of the longitudinal component of LG beams in the angular spectrum representation. Furthermore, by superposing two LG beams to form vector vortex beams (VVBs), the SOI term is preserved as $\Delta J = (\sigma' + \ell') - (\sigma + \ell)$ in the amplitude of the longitudinal field. This quantity governs the variation of the relative phase along the azimuthal angle. Thus, when $\Delta J = 0$, the azimuthal structure vanishes, and the interference is fully controlled by the relative phase between the two LG beams, leading to either complete constructive or destructive outcomes.

In Chapter 3, Fermi's golden rule is employed to evaluate the transition rates of excitons, focusing on valley-mixed bright excitons (BXs) and gray excitons (GXs) under VVB excitation. Due to the approximately isotropic momentum-space distribution and out-of-plane orientation of the GX transition dipole, the transition rate directly reflects the longitudinal component of the VVB. As a result, selective excitation of GXs is achieved under the condition $\Delta J = 0$, corresponding to complete switching of the longitudinal field. On the other hand, valley-mixed BXs exhibits behavior similar to GXs. However, the condition $\Delta J = 0$ corresponds to matching between the BX transition dipole and the light polarization. Although selective excitation of valley-mixed BXs is theoretically possible, the few-meV valley splitting poses challenges for experimental verification.

In Chapter 4, first–order time–dependent perturbation theory is applied to construct a superposition of finite-momentum exciton (SFME) states, with the aim of determining the static envelope function of the exciton wave packet and examining the resulting spatial patterns. It is found that a global phase in valley-mixed BXs modifies the resulting envelope function, necessitating an explicit treatment of the center–of–mass momentum ($\boldsymbol{Q}$)–dependent phase structure of the exciton wave function. The extraction, based on zeroth-order and related approximations, is successful for valley excitons. In contrast, valley-mixed BXs introduce additional phase complexity originating from their internal structure. Simulations for valley excitons further demonstrate that phase factors play a crucial role in shaping the static envelope function.

Future work may extend this framework to other VVB configurations and incorporate full exciton wave functions. The fascinating interplay between structured light and matter enables further control of exciton dynamics, with potential applications in valleytronics and optoelectronic systems.

References

[1] Nicolas Roisin et al. "Phonon-limited mobility for electrons and holes in highly-strained silicon". In: *npj Computational Materials* 10.1 (2024), p. 242. DOI: 10.1038/s41524-024-01425-0.

[2] Aaron Bostwick et al. "Quasiparticle dynamics in graphene". In: *Nature Physics* 3.1 (2007), pp. 36–40. DOI: 10.1038/nphys477.

[3] Andrea Splendiani et al. "Emerging Photoluminescence in Monolayer MoS2". In: *Nano Letters* 10.4 (2010), pp. 1271–1275. DOI: 10.1021/nl903868w.

[4] Alexander Krasnok and Andrea Alù. "Valley-Selective Response of Nanostructures Coupled to 2D Transition-Metal Dichalcogenides". In: *Applied Sciences* 8.7 (2018), p. 1157. DOI: 10.3390/app8071157.

[5] Guan-Hao Peng et al. "Distinctive Signatures of the Spin- and Momentum-Forbidden Dark Exciton States in the Photoluminescence of Strained WSe2 Monolayers under Thermalization". In: *Nano Letters* 19.4 (2019), pp. 2299–2312. DOI: 10.1021/acs.nanolett.8b04786.

[6] Guan-Hao Peng et al. "Tailoring the superposition of finite-momentum valley exciton states in transition-metal dichalcogenide monolayers by using polarized twisted light". In: *Physical Review B* 106.15 (2022), p. 155304. DOI: 10.1103/PhysRevB.106.155304.

[7] Lorenz Maximilian Schneider et al. "Direct Measurement of the Radiative Pattern of Bright and Dark Excitons and Exciton Complexes in Encapsulated Tungsten Diselenide". In: *Scientific Reports* 10.1 (2020), p. 8091. DOI: 10.1038/s41598-020-64838-z.

[8] J. H. Poynting. "The Wave Motion of a Revolving Shaft, and a Suggestion as to the Angular Momentum in a Beam of Circularly Polarised Light". In: *Proceedings of the Royal Society of London A* 82.557 (1909), pp. 560–567.

[9] Richard A. Beth. "Mechanical Detection and Measurement of the Angular Momentum of Light". In: *Physical Review* 50.2 (1936), pp. 115–125. DOI: 10.1103/PhysRev.50.115.

[10] L. Allen et al. "Orbital angular momentum of light and the transformation of Laguerre-Gaussian laser modes". In: *Physical Review A* 45.11 (1992), pp. 8185–8189. DOI: 10.1103/PhysRevA.45.8185.

[11] J. M. Jauch and F. Rohrlich. *The theory of photons and electrons: The relativistic quantum field theory of charged particles with spin one-half*. 2nd ed. Berlin: Springer-Verlag, 1976. DOI: 10.1007/978-3-642-80951-4.

[12] Andrew Forbes, Light Mkhumbuza, and Liang Feng. "Orbital angular momentum lasers". In: *Nature Reviews Physics* 6.6 (2024), pp. 352–364. DOI: 10.1038/s42254-024-00715-2.

[13] Manuel Erhard et al. "Twisted photons: new quantum perspectives in high dimensions". In: *Light: Science & Applications* 7.3 (2018), p. 17146. DOI: 10.1038/lsa.2017.146.

[14] H. He et al. “Direct Observation of Transfer of Angular Momentum to Absorptive Particles from a Laser Beam with a Phase Singularity”. In: *Physical Review Letters* 75.5 (1995), pp. 826–829. DOI: 10.1103/PhysRevLett.75.826.

[15] L. Paterson et al. “Controlled Rotation of Optically Trapped Microscopic Particles”. In: *Science* 292.5518 (2001), pp. 912–914. DOI: 10.1126/science.1058591.

[16] Alexander B. Stilgoe, Timo A. Nieminen, and Halina Rubinsztein-Dunlop. “Controlled transfer of transverse orbital angular momentum to optically trapped birefringent microparticles”. In: *Nature Photonics* 16.5 (2022), pp. 346–351. DOI: 10.1038/s41566-022-00983-3.

[17] Graham Gibson et al. “Free-space information transfer using light beams carrying orbital angular momentum”. In: *Optics Express* 12.22 (2004), pp. 5448–5456. DOI: 10.1364/OPEX.12.005448.

[18] Jian Wang et al. “Terabit free-space data transmission employing orbital angular momentum multiplexing”. In: *Nature Photonics* 6.7 (2012), pp. 488–496. DOI: 10.1038/nphoton.2012.138.

[19] Nenad Bozinovic et al. “Terabit-Scale Orbital Angular Momentum Mode Division Multiplexing in Fibers”. In: *Science* 340.6140 (2013), pp. 1545–1548. DOI: 10.1126/science.1237861.

[20] Giuseppe Vallone et al. “Free-Space Quantum Key Distribution by Rotation-Invariant Twisted Photons”. In: *Physical Review Letters* 113.6 (2014), p. 060503. DOI: 10.1103/PhysRevLett.113.060503.

[21] Alicia Sit et al. “High-dimensional intracity quantum cryptography with structured photons”. In: *Optica* 4.9 (2017), pp. 1006–1010. DOI: 10.1364/OPTICA.4.001006.

[22] Fabrizio Tamburini et al. “Twisting of light around rotating black holes”. In: *Nature Physics* 7.3 (2011), pp. 195–197. DOI: 10.1038/nphys1907.

[23] Dong-Sheng Ding et al. “Single-photon-level quantum image memory based on cold atomic ensembles”. In: *Nature Communications* 4.1 (2013), p. 2527. DOI: 10.1038/ncomms3527.

[24] A. Nicolas et al. “A quantum memory for orbital angular momentum photonic qubits”. In: *Nature Photonics* 8.3 (2014), pp. 234–238. DOI: 10.1038/nphoton.2013.355.

[25] Lluis Torner, Juan P. Torres, and Silvia Carrasco. “Digital spiral imaging”. In: *Optics Express* 13.3 (2005), pp. 873–881. DOI: 10.1364/OPEX.13.000873.

[26] Néstor Uribe-Patarroyo et al. “Object Identification Using Correlated Orbital Angular Momentum States”. In: *Physical Review Letters* 110.4 (2013), p. 043601. DOI: 10.1103/PhysRevLett.110.043601.

[27] Miles Padgett and Richard Bowman. “Tweezers with a twist”. In: *Nature Photonics* 5.6 (2011), pp. 343–348. DOI: 10.1038/nphoton.2011.81.

[28] K. Y. Bliokh et al. “Spin–orbit interactions of light”. In: *Nature Photonics* 9.12 (2015), pp. 796–808. DOI: 10.1038/nphoton.2015.201.

[29] Timo A. Nieminen et al. "Angular momentum of a strongly focused Gaussian beam". In: *Journal of Optics A: Pure and Applied Optics* 10.11 (2008), p. 115005. DOI: 10.1088/1464-4258/10/11/115005.

[30] Oscar Javier Gomez Sanchez et al. "Enhanced Photo-excitation and Angular-Momentum Imprint of Gray Excitons in WSe2 Monolayers by Spin–Orbit-Coupled Vector Vortex Beams". In: *ACS Nano* 18.17 (2024), pp. 11425–11437. DOI: 10.1021/acsnano.4c01881.

[31] Konstantin Y. Bliokh and Franco Nori. "Transverse and longitudinal angular momenta of light". In: *Physics Reports* 592 (2015), pp. 1–38. DOI: 10.1016/j.physrep.2015.06.003.

[32] Stephen M. Barnett and L. Allen. "Orbital angular momentum and nonparaxial light beams". In: *Optics Communications* 110.5 (1994), pp. 670–678. DOI: 10.1016/0030-4018(94)90269-0.

[33] David L. Andrews and Mohamed Babiker, eds. *The Angular Momentum of Light*. Cambridge: Cambridge University Press, 2012. DOI: 10.1017/CBO9780511795213.

[34] Safa O. Kasap and Ravindra Kumar Sinha. *Optoelectronics and Photonics: Principles and Practices*. 2nd ed. Boston: Pearson, 2013. ISBN: 9780273774174.

[35] Frank L. Pedrotti, Leno M. Pedrotti, and Leno S. Pedrotti. *Introduction to Optics*. 3rd ed. Cambridge: Cambridge University Press, 2017.

[36] L. C. Dávila Romero, D. L. Andrews, and M. Babiker. "A quantum electrodynamics framework for the nonlinear optics of twisted beams". In: *Journal of Optics B: Quantum and Semiclassical Optics* 4.2 (2002), S66–S72. DOI: 10.1088/1464-4266/4/2/370.

[37] Kayn A. Forbes, Dale Green, and Garth A. Jones. "Relevance of longitudinal fields of paraxial optical vortices". In: *Journal of Optics* 23.7 (2021), p. 075401. DOI: 10.1088/2040-8986/abff96.

[38] V. V. Klimov et al. "Mapping of focused Laguerre-Gauss beams: The interplay between spin and orbital angular momentum and its dependence on detector characteristics". In: *Physical Review A* 85.5 (2012), p. 053834. DOI: 10.1103/PhysRevA.85.053834.

[39] Shojiro Nemoto. "Nonparaxial Gaussian beams". In: *Applied Optics* 29.13 (1990), pp. 1940–1946. DOI: 10.1364/AO.29.001940.

[40] Kristan Bryan Simbulan et al. "Selective Photoexcitation of Finite-Momentum Excitons in Monolayer MoS2 by Twisted Light". In: *ACS Nano* 15.2 (2021), pp. 3481–3489. DOI: 10.1021/acsnano.0c10823.

[41] David J. Griffiths and Darrell F. Schroeter. *Introduction to Quantum Mechanics*. 3rd ed. Cambridge: Cambridge University Press, 2018.

[42] Lorenz Maximilian Schneider et al. "Optical dispersion of valley-hybridised coherent excitons with momentum-dependent valley polarisation in monolayer semiconductor". In: *2D Materials* 8.1 (2021), p. 015009. DOI: 10.1088/2053-1583/abbf94.

[43] David H. McIntyre, Janet Tate, and Chad A. Staton. *Quantum Mechanics: A Paradigms Approach*. Boston: Pearson, 2012. ISBN: 978-0321765796.

[44] Robin Bajaj et al. "Symmetries in zero and finite center-of-mass momenta excitons". In: *Physical Review B* 112 (2025), p. 245127. DOI: 10.1103/hv2z-xxt4.

[45] Jonah B. Haber et al. "Maximally localized exciton Wannier functions for solids". In: *Physical Review B* 108 (2023), p. 125118. DOI: 10.1103/PhysRevB.108.125118.

[46] Diana Y. Qiu, Ting Cao, and Steven G. Louie. "Nonanalyticity, Valley Quantum Phases, and Lightlike Exciton Dispersion in Monolayer Transition Metal Dichalcogenides: Theory and First-Principles Calculations". In: *Physical Review Letters* 115.17 (Oct. 2015), p. 176801. DOI: 10.1103/PhysRevLett.115.176801.

[47] D. G. W. Parfitt and M. E. Portnoi. "The Two-Dimensional Hydrogen Atom Revisited". In: *Journal of Mathematical Physics* 43.10 (Oct. 2002), pp. 4681–4691. DOI: 10.1063/1.1503868.

Appendix

A.1 Wave Equations from Maxwell's Equations in Free Space

In classical electrodynamics, Maxwell's equations in vacuum are given by

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r}, t) = \frac{\rho}{\epsilon_0} = 0, \tag{A.1.1}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r}, t) = 0, \tag{A.1.2}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r}, t) = -\frac{\partial \boldsymbol{B}(\boldsymbol{r}, t)}{\partial t}, \tag{A.1.3}$$

$$\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r}, t) = \mu_0 \mathbf{J} + \frac{1}{c^2}\frac{\partial \boldsymbol{E}(\boldsymbol{r}, t)}{\partial t} = \frac{1}{c^2}\frac{\partial \boldsymbol{E}(\boldsymbol{r}, t)}{\partial t}. \tag{A.1.4}$$

Here, $\boldsymbol{\nabla} \equiv \hat{\boldsymbol{x}}\frac{\partial}{\partial x} + \hat{\boldsymbol{y}}\frac{\partial}{\partial y} + \hat{\boldsymbol{z}}\frac{\partial}{\partial z}$ denotes the spatial gradient operator in Cartesian coordinates. To reformulate the equations in terms of electromagnetic potentials, we first note that Eq. (A.1.2) implies that the magnetic field can be expressed as the curl of a vector potential $\boldsymbol{A}(\boldsymbol{r}, t)$,

$$\boldsymbol{B}(\boldsymbol{r}, t) = \boldsymbol{\nabla} \times \boldsymbol{A}(\boldsymbol{r}, t). \tag{A.1.5}$$

Substituting Eq. (A.1.5) into Faraday's law (Eq. (A.1.3)) gives a curl-free expression, which allows the introduction of a scalar potential $\phi(\boldsymbol{r}, t)$ such that

$$\boldsymbol{E}(\boldsymbol{r}, t) = -\boldsymbol{\nabla}\phi(\boldsymbol{r}, t) - \frac{\partial \boldsymbol{A}(\boldsymbol{r}, t)}{\partial t}. \tag{A.1.6}$$

This representation automatically satisfies Eq. (A.1.2) and Eq. (A.1.3). Finally, Gauss's law (Eq. (A.1.1)) and Ampère−Maxwell law (Eq. (A.1.4)) can be rewritten as coupled wave equations for the scalar and vector potentials:

$$\nabla^2\phi(\boldsymbol{r}, t) + \frac{\partial}{\partial t}\big(\boldsymbol{\nabla} \cdot \boldsymbol{A}(\boldsymbol{r}, t)\big) = 0, \tag{A.1.7}$$

$$\nabla^2\boldsymbol{A}(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2 \boldsymbol{A}(\boldsymbol{r}, t)}{\partial t^2} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \boldsymbol{A}(\boldsymbol{r}, t) + \frac{1}{c^2}\frac{\partial \phi(\boldsymbol{r}, t)}{\partial t}\right) = 0. \tag{A.1.8}$$

A.2 Scalar Helmholtz Equation

We begin in the Lorenz gauge and assume a monochromatic-wave ansatz of the form $\boldsymbol{A}^L(\boldsymbol{r}, t) = \mathrm{Re}\big[\boldsymbol{A}^L(\boldsymbol{r})e^{-i\omega t}\big]$. Substituting this expression into Eq. (2.8) yields the vector Helmholtz equation

$$\nabla^2\boldsymbol{A}^L(\boldsymbol{r}) + q_0^2\boldsymbol{A}^L(\boldsymbol{r}) = 0. \tag{A.2.1}$$

We then impose a transverse-field ansatz, $\boldsymbol{A}^L(\boldsymbol{r}) = \hat{\boldsymbol{\varepsilon}}\, A^L(\boldsymbol{r})$, where $\hat{\boldsymbol{\varepsilon}}$ is the polarization unit vector. Under this assumption, Eq. (A.2.1) reduces to a scalar equation for the amplitude $A^L(\boldsymbol{r})$,

$$\nabla^2 A^L(\boldsymbol{r}) + q_0^2 A^L(\boldsymbol{r}) = 0, \tag{A.2.2}$$

which is the scalar Helmholtz equation.

A.3 Valley Excitons with Zero Center-of-Mass Momentum

In this appendix, we derive the relation between valley exciton wave functions with zero center-of-mass momentum at the high-symmetry valleys K and K'. Using the Bloch theorem, the corresponding exciton wave functions can be written as

$$\begin{aligned} U^{K,\mathbf{0}}(\boldsymbol{R}_c, \boldsymbol{r}) &\approx \psi_{c,\boldsymbol{K}}(\boldsymbol{R}_c)\psi^*_{v,\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r}) &&= u_{c,\boldsymbol{K}}(\boldsymbol{R}_c)u^*_{v,\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r})e^{i\boldsymbol{K}\cdot\boldsymbol{r}}, \\ U^{K',\mathbf{0}}(\boldsymbol{R}_c, \boldsymbol{r}) &\approx \psi_{c,-\boldsymbol{K}}(\boldsymbol{R}_c)\psi^*_{v,-\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r}) &&= u_{c,-\boldsymbol{K}}(\boldsymbol{R}_c)u^*_{v,-\boldsymbol{K}}(\boldsymbol{R}_c - \boldsymbol{r})e^{-i\boldsymbol{K}\cdot\boldsymbol{r}}, \end{aligned} \tag{A.3.1}$$

Here, the vectors $\boldsymbol{K}$ and $\boldsymbol{K'}$ denote the momentum-space position vectors of the K and K' valleys in reciprocal space, respectively. The single-particle Bloch wave functions satisfy[1]

$$\psi^*_{n,-\boldsymbol{K}}(\boldsymbol{r'}) = \psi_{n,\boldsymbol{K}}(\boldsymbol{r'}). \tag{A.3.2}$$

Using the Bloch form of the single-particle wave function,

$$\psi_{n,\boldsymbol{k}}(\boldsymbol{r'}) = u_{n,\boldsymbol{k}}(\boldsymbol{r'})e^{i\boldsymbol{k}\cdot\boldsymbol{r'}}, \tag{A.3.3}$$

Eq. (A.3.2) gives

$$u^*_{n,-\boldsymbol{K}}(\boldsymbol{r'}) = u_{n,\boldsymbol{K}}(\boldsymbol{r'}). \tag{A.3.4}$$

Substituting this relation into Eq. (A.3.1), we obtain

$$U^{K,\mathbf{0}}(\boldsymbol{R}_c, \boldsymbol{r}) = U^{K',\mathbf{0}*}(\boldsymbol{R}_c, \boldsymbol{r}) \tag{A.3.5}$$

which establishes the complex-conjugation relation between the valley exciton wave functions at $\boldsymbol{K}$ and $\boldsymbol{K'}$.

[1] The vector $\boldsymbol{r'}$ denotes a general position vector and may represent the weighted-average, relative, electron, or hole coordinate.